

Classification and Management of Chronic Kidney Disease

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CHAPTER OUTLINE

INTRODUCTION, 1946

CLASSIFICATION OF CHRONIC KIDNEY DISEASE, 1947

MECHANISMS OF DISEASE PROGRESSION AND THE RATIONALE FOR INTERVENTIONS TO ACHIEVE RENOPROTECTION, 1947

INTERVENTIONS FOR SLOWING PROGRESSION OF CHRONIC KIDNEY DISEASE, 1948

INTERVENTIONS TO REDUCE CARDIOVASCULAR RISK ASSOCIATED WITH CHRONIC KIDNEY DISEASE, 1971

INTERVENTIONS TO MANAGE COMPLICATIONS OF CHRONIC KIDNEY DISEASE, 1972

A STEPPED CARE APPROACH TO CHRONIC KIDNEY DISEASE, 1973

KEY POINTS

- The 2012 Kidney Disease Improving Global Outcomes (KDIGO) classification system for chronic kidney disease (CKD) provides simple risk stratification and serves as a convenient framework for guiding management through different stages.
- Lifestyle measures including smoking cessation, weight loss, and dietary sodium restriction should form part of any strategy to achieve renoprotection and reduce cardiovascular risk in persons with CKD.
- Antihypertensive therapy with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers as “first line” in persons with albuminuria is the mainstay of treatment to achieve optimal renoprotection.
- Evidence from The Systolic Blood Pressure Intervention Trial (SPRINT) indicates that a lower target for the treatment of systolic blood pressure may afford improved cardiovascular risk reduction but may also be associated with some increase in risk of acute kidney injury and hyperkalemia. The risks versus benefits of a lower blood pressure target should be evaluated in each individual.
- To achieve optimal renoprotection careful attention should be paid to addressing all modifiable aspects of CKD to achieve target blood pressure, minimize proteinuria, and slow glomerular filtration rate decline.
- All persons with CKD should be regarded as having a high risk of cardiovascular disease and management offered to reduce this risk, including treatment with a statin.

INTRODUCTION

Chronic kidney disease (CKD) is a collective term covering all primary disease processes that result in structural or functional kidney abnormalities, or both, persisting for at least 3 months. Abnormal urinalysis results identifying proteinuria or hematuria and abnormal kidney structure or histologic features, with or without a decreased glomerular

filtration rate (GFR), are the defining manifestations.¹ The rationale for having a single term that encompasses a wide range of kidney pathologies is that damage to the kidneys results in shared pathophysiological features and that many of the interventions to slow the progression of CKD or reduce the associated cardiovascular risk should be applied regardless of the etiology of the CKD. The aforementioned formal definition of CKD was proposed relatively recently in 2002 and has had a profound impact on the specialty

of nephrology. Prior to this it was difficult to compare the results of epidemiological and clinical studies because each used a different definition. In addition, there was no unified terminology and several imprecise terms such as chronic renal disease, impairment, insufficiency, and failure were used by different authors. Epidemiological research based on the unified definition has revealed that CKD is common, affecting on average 13.4% of the adult population globally,² and CKD has been identified as a leading cause of death in global burden of disease studies,³ though it should be noted that most epidemiological studies rely on only a single abnormal GFR value for the diagnosis of CKD and therefore probably overestimate the true prevalence. Nevertheless, nephrology has changed from a specialty that focused largely on the treatment of rare and advanced kidney diseases as well as renal replacement therapy (RRT) to one that is now concerned with a condition, CKD, that affects a substantial proportion of the general population. Of necessity, this has also resulted in the need for healthcare workers who are not nephrologists to gain knowledge of CKD and its management. Importantly, most of the treatments required for CKD are relatively cheap and simple to apply and can therefore be delivered in healthcare settings where diagnostic services are limited and provision of RRT is inadequate. Thus education of healthcare workers on the management of CKD has the potential to reduce the burden of disease and the need for RRT without the need for sophisticated diagnostics or expensive drugs. This chapter will review in detail the classification of CKD and the evidence-based treatments that should be offered to persons with CKD regardless of the underlying kidney pathology. Within this context, there are four broad aims:

1. Attenuate GFR decline and thus prevent or delay the need for dialysis. It is perhaps self-evident, but the earlier that CKD progression can be halted, the greater is the possibility of maintaining kidney function as close to normal as possible.
2. Prevent premature cardiovascular death at all stages of CKD.
3. Recognize and manage complications of CKD as they arise, particularly in stages 4 and 5.
4. Timely preparation for RRT or conservative and palliative care.

Because the pathophysiology associated with CKD changes as the condition progresses, we also propose a stepped care approach and consider how primary and secondary care clinicians should cooperate to provide optimal care for the large number of persons now identified to have CKD.

CLASSIFICATION OF CHRONIC KIDNEY DISEASE

The original classification proposed in 2002 subdivided CKD into five stages according to the GFR, reflecting the observation that in the majority of cases, CKD progresses slowly through the stages before reaching end-stage kidney disease (ESKD). In 2008, a National Institute for Health and Care Excellence (NICE) guideline proposed that stage 3 (GFR 59–30 mL/min/1.73 m²) be subdivided into stages 3a (GFR 59–45 mL/min/1.73 m²) and 3b (GFR 44–30 mL/min/1.73 m²), reflecting a broad consensus

that there are important differences in clinical aspects and prognosis between these groups. The CKD staging system has two important implications: First, it suggests that if CKD is detected at an early stage, intervention may be possible to prevent or slow progression to more advanced stages.⁴ Second, it reflects the observation that as GFR declines, the risk profile of persons and associated complications changes. Thus the staging system provides a useful framework for structuring therapy and prioritizing interventions to produce a comprehensive strategy for the management of CKD. Although universally adopted, this classification system was limited by not reflecting the risk status of those classified, a feature that is present in the many other disease classification systems. Following extensive epidemiological investigations that identified reduced GFR and albuminuria as independent risk factors for adverse outcomes, the CKD classification system was revised by Kidney Disease Improving Global Outcomes (KDIGO) in 2012. The GFR stages were preserved, though now termed categories G1 to G5, and new categories were added for albuminuria; categories A1 to A3, corresponding to the previously used descriptions of “normoalbuminuria,” “microalbuminuria,” and “macroalbuminuria.” In addition, it was emphasized that the etiology of the CKD should be included in the classification, giving rise to the term “CGA” classification (cause, GFR, albuminuria; Fig. 59.1).^{5,6} According to this system, a person with immunoglobulin A (IgA) nephropathy, a GFR of 34 mL/min/1.73 m², and a urine albumin-to-creatinine ratio (UACR) of 367 mg/g would be classified as “IgA nephropathy, category G3b A3”. Implementation of the CGA classification has been greatly aided by the global adoption of equations to derive an estimated GFR (eGFR) from the serum creatinine concentration and use of UACR on a random sample to quantitate albuminuria (see Chapter 23). Importantly, in this classification system each category reflects a risk status for multiple adverse outcomes including ESKD, progression of CKD, cardiovascular events (CVEs), and all-cause mortality as indicated in the accompanying heat map. Thus the category is useful to guide the therapeutic approach at each stage and prompt referral for specialist care when indicated.

Despite its global adoption, some commentators have questioned the utility of the CKD definition and classification, suggesting that the high prevalence of CKD reflects a normal aging process and pointing out that mildly reduced GFR (category G3a) in the absence of albuminuria is associated with minimal increase in risk, particularly in older persons.^{7,8} Various solutions have been proposed to address this issue, including the introduction of an age-adjusted GFR threshold for defining CKD or a modification of the classification for different age groups, but to date no consensus has been reached and the KDIGO classification is used worldwide.

MECHANISMS OF DISEASE PROGRESSION AND THE RATIONALE FOR INTERVENTIONS TO ACHIEVE RENOPROTECTION

It has long been appreciated that regardless of the primary renal disease, kidney damage tends to progress toward ESKD, especially if more than half of glomeruli have been lost. This suggests that a common pathway of mechanisms may promote

Prognosis of CKD by GFR and Albuminuria categories: KDIGO 2012

				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (mL/min/1.73 m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

Green: low risk (if no other markers of kidney disease, no CKD)
 Yellow: moderately increased risk
 Orange: high risk
 Red: very high risk

Fig. 59.1 Current classification system and nomenclature proposed by the Kidney Disease Improving Global Outcomes (KDIGO) in 2012. Chronic kidney disease (CKD) is defined as abnormalities of kidney structure or function, present for 3 months, with implications for health. CKD is classified based on cause, glomerular filtration rate (GFR) category, and albuminuria category. (From Kidney Disease: Improving Global Outcomes CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int.* 2013;3(Suppl):1–150. With permission.)

kidney damage and establish a vicious cycle of nephron loss. In research efforts since the 1960s, investigators have identified glomerular hemodynamic factors (glomerular hypertension and hyperfiltration), multiple effects of angiotensin II, proteinuria, and proinflammatory and profibrotic molecules as key elements of this pathway (discussed in detail in Chapter 51). A common pathway underlies progressive kidney damage from kidney diseases of diverse causes, and recognition of this pathway has been vital in informing strategies to achieve renoprotection. Thus the interventions to slow CKD progression discussed in the following sections are each aimed at attenuating mechanisms of progression. In view of the redundancy that is characteristic of most biologic systems, it has become clear that to achieve optimal renoprotection, attempts should be made to inhibit the vicious cycle of common pathway mechanisms at multiple points (Fig. 59.2).

INTERVENTIONS FOR SLOWING PROGRESSION OF CHRONIC KIDNEY DISEASE

LIFESTYLE INTERVENTIONS

All guidelines support healthy lifestyle advice for persons with CKD, citing the substantial literature that exists largely

for the general population. A major issue for nephrologists and physicians managing CKD is that the available randomized controlled trials (RCTs) of the effect of lifestyle interventions rarely enrolled persons with CKD. Moreover, lifestyle changes require considerable effort from persons and may take years to be effective. In the prospective observational Chronic Renal Insufficiency Cohort (CRIC) study, regular physical activity, nonsmoking, and a body mass index (BMI) 25 kg/m² or higher were associated with lower risk of adverse outcomes, including a 50% decrease in eGFR or ESKD, atherosclerotic events, and all-cause mortality,⁹ but there are potential biases in such observational studies because cardiovascular diseases and CKD may themselves result in low physical activity (reverse causality). Evidence does exist for the value of lifestyle intervention with regard to treating hypertension and preventing CVDs. Although specific studies of lifestyle interventions in persons with CKD are still lacking, it is reasonable to expect that a similar relative (and greater absolute) benefit might be gained.

SMOKING CESSATION

Tobacco use is the most common cause of avoidable cardiovascular mortality worldwide; therefore, not surprisingly, smoking cessation is one of the most promoted methods in management of cardiovascular risk in the general population and in persons with CKD. Evidence that smoking cessation

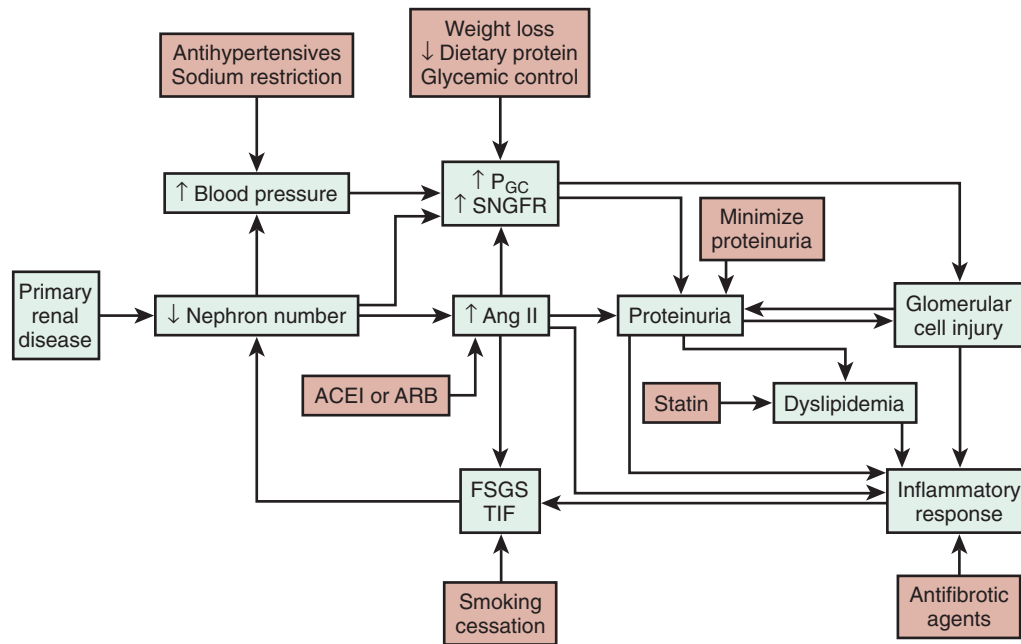


Fig. 59.2 A common pathway of mechanisms that result in a vicious circle of nephron loss in chronic kidney disease (CKD). Interventions (in red) for achieving renoprotection are directed at inhibiting the common pathway at multiple points to slow CKD progression. ACEI, Angiotensin-converting enzyme inhibitor; Ang II, angiotensin II; ARB, angiotensin receptor blocker; FSGS, focal segmental glomerulosclerosis; P_{Gc} , glomerular capillary hydraulic pressure; SNGFR, single nephron glomerular filtration rate; TIF, tubulointerstitial fibrosis.

helps prevent CKD progression is emerging.¹⁰ In the Multiple Risk Factor Intervention Trial (MRFIT), smoking was significantly associated with an increased risk for ESKD¹¹; in the Prevention of Renal and Vascular End-stage Disease (PREVEND) study, the rate of urine albumin excretion was correlated with the number of cigarettes smoked.¹² Smoking has been identified as a risk factor for the development of microalbuminuria and overt proteinuria and for the progression of CKD in persons with type 1 and type 2 diabetes.^{13–15} In a large Swedish study, the risk for disease-specific types of CKD among smokers was compared with that among people who had never smoked. Overall, the association was modest, but an important finding was that the risk increased with high daily consumption (>20 cigarettes/day), long duration (>40 years), and a high cumulative “dose” (>30 pack-years) in comparison to participants who had never smoked. Smoking increased risk most strongly for persons with CKD classified as nephrosclerosis and also glomerulonephritis.¹⁶ In the Jackson Heart Study, smoking was associated with a higher risk of rapid GFR decline (>30% decline from baseline) in a dose-dependent manner¹⁷ and in a combined analysis of five large US cohort studies ($n = 954,029$), smoking was associated with a doubling in the risk of death from ESKD.¹⁸ Finally, a meta-analysis of 15 community-based cohort studies observed an increase in the risk of developing CKD in former (hazard ratio [HR], 1.15; 95% confidence interval [CI], 1.08–1.23) and current smokers (HR, 1.34; 95% CI, 1.23–1.47) as well as an increased risk of ESKD in former (HR, 1.44; 95% CI, 1.00–2.09) and current smokers (HR, 1.91; 95% CI, 1.39–2.64), though smoking was not associated with the development of proteinuria.¹⁹

Smoking has been described as a risk factor for progression in various forms of nondiabetic CKD. Of persons with adult polycystic kidney disease, immunoglobulin A (IgA) nephropa-

thy, and lupus nephritis, those who were smokers had a substantially increased risk of progression to ESKD in comparison with nonsmokers.^{15,20} Two large cohort studies have reported detailed data on the impact of smoking on CKD progression. In the CRIC study, smoking was associated with increased all-cause mortality as well as the combined endpoint of all-cause mortality and CKD progression, though not with CKD progression alone.²¹ In an analysis of data from the Study of Heart and Renal Protection (SHARP) trial, current and former smoking were associated with higher levels of proteinuria but not with CKD progression during a median follow-up 4.9 years. Smoking was associated with significantly increased risks of CVDs, cancer, and all-cause mortality.²² RCTs on the effect of smoking cessation on CKD progression have yet to be published, and few prospective data are available, but in one study of persons with diabetes, smoking cessation was associated with less progression to macroalbuminuria (proteinuria) and a slower rate of GFR decline in comparison to persons who continued smoking.²³ In addition, the intermediate risk for adverse outcomes reported for ex-smokers versus current smokers in several of the aforementioned studies provides indirect evidence of benefit from smoking cessation. Further prospective studies are required, but published evidence strongly supports smoking cessation as an intervention to reduce the incidence and progression of CKD.

Pharmacotherapy to assist in smoking cessation is now well established. A meta-analysis of RCTs in the general population (69 trials involving a total of 32,908 persons) showed that varenicline, bupropion, and five nicotine replacement therapies were all more efficacious than placebo at promoting smoking abstinence at 6 and 12 months. In comparison with persons taking placebo, persons treated with these agents were 1.5 to 2.5 times more likely to quit smoking, depending

on the specific agent.²⁴ Indeed, combining varenicline and a nicotine replacement patch achieved an impressive higher continuous abstinence rate at 12 weeks (55.4% vs. 40.9%) and 6-month point prevalence abstinence rate of 65.1% versus 46.7% compared with varenicline treatment alone.²⁵ An important finding is that people who smoke are more likely to stop smoking if offered a combination of interventions, such as behavioral support and pharmacotherapy. Multicomponent interventions are now part of many public health guidelines. We see no reason to exclude people with CKD from this advice.

WEIGHT LOSS

Obesity is the dominant risk factor for type 2 diabetes and is also a major risk factor for hypertension and progression of CKD. Evidence linking the metabolic syndrome and CKD has emerged; each element of the metabolic syndrome is associated with increased prevalence of CKD and microalbuminuria. In animal models (obese Zucker rats with type 2 diabetes), early progressive podocyte damage and macrophage infiltration are associated with hyperlipidemia and antedate both the development of glomerulosclerosis and tubulointerstitial damage.^{26,27} In humans, there was a graded relationship between the number of components of the metabolic syndrome present and the corresponding prevalence of CKD or microalbuminuria.²⁸

Epidemiologic studies have identified obesity as a risk factor for CKD,^{29,30} and in one study, obesity was an independent risk factor for progression of IgA nephropathy.³¹ Furthermore, the largest such study to assess risk of CKD in association with obesity demonstrated a very strong biologic gradient: Increasing BMI was associated with increasing risk of ESKD. In comparison to persons with an “ideal” BMI (18.5–24.9 kg/m²), the relative risk (RR) of ESKD was 3.6-fold for those with a BMI of 30 to 34.9 kg/m², sixfold for those with BMI of 35 to 39.9 kg/m², and sevenfold for those with BMI of 40 kg/m² or higher. Controlling for baseline blood pressure and presence of diabetes attenuated the associations, but the gradient between increasing body size and ESKD risk remained strong.³²

Because these components are major factors in the initiation and progression of CKD, respectively, persons with CKD would probably benefit from weight loss and reversal of the features of the metabolic syndrome, as observed in the general population.³³ Results of early studies support this assertion, inasmuch as weight loss in humans with obesity demonstrated reversal of glomerular hyperfiltration and albuminuria.³⁴

Moreover, there is evidence that weight loss of as little as 10 lbs (4.5 kg) reduces blood pressure, prevents hypertension, or does both in a large proportion of overweight persons, although the ideal is to maintain normal body weight.³⁵ In the Framingham Heart Study, weight loss of 5 lbs (2.25 kg) or more was associated with reductions in cardiovascular risk of about 40%³⁶ for both men and women, and thus should be a clear goal for persons with CKD who are overweight. It also appears that the degree of weight loss, regardless of method (lifestyle changes or bariatric surgery), dictates the benefits of lowering of blood pressure and reduction in glycemic markers.³⁷ However, according to longer-term studies of lifestyle modification and of persons after bariatric surgery, the blood pressure–lowering benefits regress somewhat over time, although the vascular outcomes continue to be better than those in control groups.^{33,38,39}

Renoprotective effects associated with weight loss interventions (dietary caloric restriction, exercise, antiobesity medications, and bariatric surgery) were reported in a meta-analysis of data from 522 participants in five controlled and eight uncontrolled trials.⁴⁰ In persons undergoing intervention to lose weight, those with proteinuria had a mean reduction of urine protein levels of 1.7 g/day; even among persons with microalbuminuria, mean reduction of urinary albumin excretion was 14 mg/day. Although these reductions were modest in comparison with those in persons with overt proteinuria, results of other studies of blood pressure and glycemic control interventions on microalbuminuria suggest that they will have long-term benefit in such persons.⁴⁰ This was confirmed by a secondary analysis of the Look AHEAD trial in which 5145 overweight or obese persons with type 2 diabetes were randomly assigned to an intensive lifestyle intervention (ILI) or diabetes support and education (DSE). The trial was stopped early due to lack of any difference in the primary outcome, CVDs, but analysis of renal outcomes showed a 31% reduction in the incidence of “high-risk CKD” as defined by the KDIGO classification system (HR, 0.69; 95% CI, 55 to 0.87) in the ILI versus DSE group, mediated in part by weight loss and improvements in hypertension and glycemic control.⁴¹ A systematic review of 31 studies analyzed the effects of weight loss achieved by bariatric surgery, medication, or diet and found that in the majority of studies, weight loss was associated with reductions in proteinuria. In people with glomerular hyperfiltration, GFR tended to decrease with weight loss, and in those with reduced GFR, it tended to increase.⁴²

Increasing use of bariatric surgery to manage obesity has resulted in several studies evaluating impact on kidney function and incidence of CKD. In one long-term study of 2144 persons who underwent bariatric surgery, renal outcomes were assessed after 7 years as change in CKD risk category as defined by the KDIGO classification. Among those with moderate risk at baseline, the risk category improved in 53% and deteriorated in 5% to 8%; in the high-risk group, improvement was observed in 56% and deterioration in 3% to 10%; in the very-high-risk group, 23% improved. eGFR initially improved with a peak at 2 years and then gradually declined. Albuminuria showed large and sustained decreases in moderate (median UACR 48–14 mg/g) and high-risk groups (median UACR 326–26 mg/g).⁴³ In another study that included only persons with CKD stage 3 and 4 and propensity score matching with similarly obese persons who did not undergo bariatric surgery, eGFR was significantly higher by a mean of 9.84 mL/min/1.73 m² in the surgery group after 3 years.⁴⁴ These studies are limited by relying on eGFR to assess kidney function but the finding that benefit was sustained over several years provides reassurance that improvement was not simply an artifact due to early decrease in serum creatinine secondary to weight loss after surgery. In a meta-analysis that included 23 cohort studies and 3015 persons who underwent all forms of bariatric surgery, small but statistically significant improvements were observed in serum creatinine (mean decrease 0.08 mg/dL) and proteinuria (mean decrease 0.04 g/day).⁴⁵

Further large randomized studies of weight loss in persons with CKD are required because gastric bypass surgery is associated with renal risks including increased risk of renal calculi^{46,47} and treatment with orlistat is associated with

increased risk of acute kidney injury (AKI) as well as CKD.⁴⁸ On the basis of the available data, we recommend weight loss in obese persons with CKD through a combination of increased exercise and reduced caloric intake, and consideration of bariatric surgery if these are not successful.

DIETARY SODIUM RESTRICTION

There is substantial evidence from epidemiologic, migration, intervention, genetic, and animal studies that salt intake plays an important role in regulating blood pressure. Essential hypertension is observed primarily in societies in which the average sodium intake exceeds 100 mEq/day (2.3 g sodium or $\approx$ 6 g sodium chloride) and is rare in societies in which the average sodium intake is less than 50 mEq/day (1.2 g sodium or 3 g sodium chloride).⁴⁹ This relationship between sodium intake and population levels of hypertension is strong and consistent. Of importance is that sodium restriction produces a significant reduction in blood pressure.^{50,51} Salt (sodium chloride) intake in many countries is 9 to 12 g/day. The World Health Organization's current recommendation for adults is to reduce salt intake to 5 g/day or less.

A meta-analysis of RCTs with a duration of at least 4 weeks concluded that reducing salt intake by 3 g/day is predictive of a linear fall in blood pressure of, on average, 3.6 to 5.6 mm Hg (systolic blood pressure [SBP]) and 1.9 to 3.2 mm Hg (diastolic blood pressure [DBP]) in hypertensive persons and of 1.8 to 3.5 mm Hg (SBP) and 0.8 to 1.8 mm Hg (DBP) in normotensive participants.⁵² The effect may be doubled with a 6 g/day reduction, and estimates suggest that reducing salt intake by 9 g/day (e.g., from 12 to 3 g/day) would reduce strokes by approximately 33% and ischemic heart disease by 25%. Weight loss and reduced sodium intake are particularly beneficial in older people. In the Trial of Nonpharmacologic Interventions in the Elderly (TONE), reducing sodium intake to 80 mEq (2 g) per day reduced blood pressure over 30 months, and about 40% of participants on the low-salt diet were able to discontinue their antihypertensive medications.⁵³

In many persons with CKD, especially those with glomerular disease and severe proteinuria, hypertension behaves in a salt-sensitive manner. In one large observational study of persons without CKD at baseline, both high (>4.03 g/day of sodium) and low (<2.08 g/day of sodium) dietary sodium intake were associated with an increased incidence of CKD during follow-up of over 10 years in 3106 hypertensive participants, but no association with sodium intake was observed in those without hypertension ($n = 4871$).⁵⁴ This supports that salt sensitivity is important in the association between sodium intake and CKD. The reason for the association with low sodium intake is unknown but may be confounded by associations between sodium intake and several other nutritional factors including potassium, total calorie, fat, protein, and carbohydrate intake, though the analysis was adjusted for these factors. A systematic review of 16 small studies that investigated the association of dietary sodium intake and CKD progression concluded that marked heterogeneity between the studies precluded meta-analysis.⁵⁵ Nevertheless, the general trend observed was that increasing sodium intake is associated with worsening albuminuria. An analysis of 3757 participants with CKD from the CRIC study found that after 15,807 person-years of follow-up, the highest quartile of urinary sodium excretion (≥ 195 mmol/day) was

associated with a significantly increased risk of CKD progression (HR, 1.54; 95% CI, 1.23–1.92) and all-cause mortality (HR, 1.45; 95% CI, 1.08–1.95) compared with the lowest quartile (<117 mmol/day).⁵⁶

High dietary sodium intake has also been shown to negate the antiproteinuric effects of treatment with angiotensin-converting enzyme (ACE) inhibitors.⁵⁷ In a prospective, randomized, placebo-controlled crossover study, dietary sodium restriction increased the antihypertensive and antiproteinuric effects of therapy with angiotensin receptor blockers (ARBs), as monotherapy or in combination with a thiazide diuretic, in persons with nondiabetic CKD (Fig. 59.3).⁵⁸ Whereas the protocol aimed to restrict dietary salt intake to less than 50 mEq/day (1.2 g of sodium or 3 g of sodium chloride), these impressive results were observed with achieved sodium restriction of only 92 mEq/day.⁵⁸ Furthermore, a post hoc analysis of data from the first and second Ramipril Efficacy In Nephropathy (REIN) trials found that medium and high sodium intake were associated with significant increases in the incidence of ESKD versus low sodium intake. Each 100-mEq/g increase in 24-hour urinary sodium/creatinine excretion was associated with a 1.61-fold (95% CI, 1.15–2.24) higher risk of ESKD, independent of blood pressure.⁵⁹

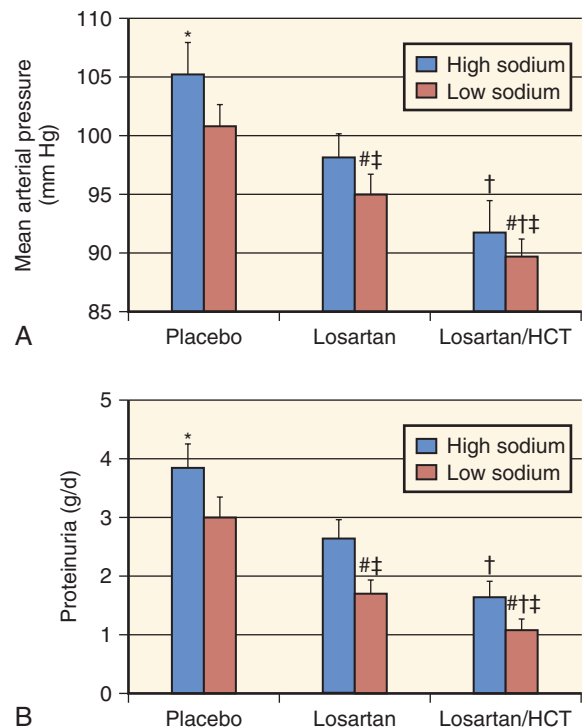


Fig. 59.3 Results of a prospective randomized crossover trial showing the effect of dietary sodium restriction on the (A) antihypertensive and (B) antiproteinuric effects of treatment with an angiotensin receptor blocker as monotherapy or in combination with a thiazide diuretic. * $P < .05$ versus all periods; # $P < .05$ versus same treatment on high-salt diet (effect of low-salt diet); † $P < .05$ versus losartan treatment on same diet [effect of hydrochlorothiazide (HCT)]; †† $P < .05$ versus placebo on same diet. (From Vogt L, Waanders F, Boomsma F, et al. Effects of dietary sodium and hydrochlorothiazide on the antiproteinuric efficacy of losartan. *J Am Soc Nephrol*. 2008;19:999–1007.)

Several small prospective studies have investigated the impact of dietary sodium restriction on blood pressure and proteinuria in persons with CKD. In a randomized placebo-controlled crossover study (achieved using low-sodium diet of 60 to 80 mmol/day plus sodium chloride tablets [120 mmol/day] or placebo) in 20 people with stage 3 or stage 4 CKD, low sodium intake was associated with an average 10/4 mm Hg reduction in blood pressure as well as reductions in albuminuria and proteinuria.⁶⁰ In a similar randomized crossover study dietary sodium restriction to less than 2 g/day for 4 weeks in persons with CKD stage 3 and 4 was associated with an average reduction in extracellular volume of 1.02 L, weight loss of 2.3 kg, and mean reduction in ambulatory SBP of 10.8 mm Hg but no improvement in albuminuria.⁶¹ By contrast, another study in persons with CKD stage 1–3 and persistent albuminuria greater than 300 mg/day despite treatment with ramipril 10 mg/day reported that dietary sodium restriction for 8 weeks was associated with a reduction in arterial pressure from a mean of 95 to 90 mm Hg and a significant decrease in albuminuria from a mean of 1060 to 717 mg/day. Creatinine clearance also decreased slightly from a mean 101 to 91 mL/min.⁶² Another study illustrated the challenges associated with achieving sustained reduction in sodium intake. In an RCT, persons with CKD assigned to a 3-month intervention of education and coaching as well as self-monitoring achieved a modest mean reduction in urinary sodium excretion of 30 mmol/day associated with a mean reduction in ambulatory DBP of 3.4 mm Hg and mean reduction in proteinuria of 0.4 g/day, but at 3 months after completion of the intervention there was no difference in urinary sodium excretion or ambulatory BP compared with controls, although proteinuria remained 0.3 g/day lower.⁶³ Further long-term randomized trials are needed to define the role of sodium restriction in renoprotective strategies, but even the incomplete evidence available supports a recommendation for moderate dietary sodium restriction to less than 5 g/day of salt in persons with CKD.

Food processing drastically changes the cationic content of natural foods, increasing sodium content and decreasing potassium content. On average, approximately 10% of dietary sodium chloride originates naturally in foods, whereas approximately 80% is the result of food processing, the remainder being discretionary (added during cooking or at the table). We advocate assessment of salt intake in individuals with CKD and advise reducing salt intake with the assistance of a dietitian to less than 5 g/day (<90 mmol/day of sodium) as recommended by KDIGO.⁵

DIETARY PROTEIN RESTRICTION

Dietary protein restriction as a renoprotective strategy is based on the notion that reducing the excretory burden on the kidneys would slow the rate of progressive injury. Accordingly, dietary protein restriction was among the first interventions proposed to slow CKD progression. Experimental studies showed that a low-protein diet (LPD) normalized glomerular hemodynamics in the remnant kidney model⁶⁴ and resulted in effective long-term renoprotection.⁶⁵ Unfortunately, clinical studies to date have failed to provide unambiguous evidence to support the use of protein restriction in human CKD. The proposal of nomenclature and diagnostic criteria for protein-energy wasting in AKI and CKD was an important step in planning future studies.⁶⁶ The suggested criteria for

diagnosing protein-energy wasting include the presence of reduced body mass (specifically muscle mass) and serum evidence of low albumin, prealbumin, or cholesterol levels.

The Modification of Diet in Renal Disease (MDRD) study was designed to provide a definite answer to the question of whether dietary protein restriction slows the progression of CKD. The study had two components: In study A, 585 persons with mostly nondiabetic CKD (GFR, 25–55 mL/min/1.73 m²) were randomly assigned to follow either a diet with “usual” protein levels (1.3 g/kg/day) or LPD (0.58 g/kg/day); in study B, 255 persons with GFRs of 13 to 24 mL/min/1.73 m² were randomly assigned to follow either LPD (0.58 g/kg/day) or “very low” protein diet (VLPD; 0.28 g/kg/day) with keto-amino acid supplementation to prevent malnutrition. After a mean follow-up period of 2.2 years, there was no difference in the rate of GFR decline in study A and only a trend toward slower decline in the VLPD group in study B.⁶⁷ Further analysis indicated, however, that the desired protein intake was not achieved in the randomized groups, and secondary analysis based on achieved dietary protein intake demonstrated that a reduction in protein intake of 0.2 g/kg/day was correlated with a 1.15-mL/min/year reduction in the rate of GFR decline, equivalent to a 29% reduction in mean rate of GFR decline.⁶⁸ In addition, a post hoc two-slope analysis, in which presumptive acute effects of dietary protein restriction were taken into account, suggested a modest long-term benefit. Long-term follow-up of the cohort in MDRD study A yielded disappointingly inconclusive results.⁶⁹ In a more recent trial, 207 well-nourished persons without diabetes and with eGFR less than 30 mL/min/1.73 m² and urine protein-to-creatinine ratio (PCR) less than 1 g/g were randomized to a vegetarian VLPD (<0.3 g/kg/day) with ketoanalog supplementation (ketoanalog diet [KD]) or standard (nonvegetarian) LPD (<0.6 g/kg/day). After 15 months the primary outcome of RRT or 50% reduction in eGFR was reached in 13% of those on KD versus 42% on LPD (adjusted HR, 0.1; 95% CI, 0.05–0.2). Those on the KD also evidenced a 3.2 mL/min/year slower rate of eGFR decline. Metabolic factors were also improved by the KD; serum bicarbonate and calcium were higher and phosphate lower. There was no change in nutritional status in either group and no adverse reactions were reported.⁷⁰ However, it is unclear what proportion of the benefit was attributable to the vegetarian VLPD versus the ketoanalog supplementation.

The effect of dietary protein restriction has been examined in several meta-analyses of smaller randomized studies. Pedrini and associates⁷¹ summarized findings on 1413 persons with nondiabetic CKD from five studies (including those from study A of the MDRD trial). LPD was associated with an RR of 0.67 (95% CI, 0.50–0.89) for ESKD or death. Similarly, among 108 persons with type 1 diabetes from five studies, LPD significantly slowed the increase in albuminuria or the decline in GFR or creatinine clearance (RR, 0.56; 95% CI, 0.40–0.77). Kasiske and colleagues⁷² pooled the results of 13 RCTs (1919 persons) and found that dietary protein restriction reduced the rate of decline in eGFR by 0.53 mL/min/year.

Larger effects were observed in participants with diabetes, and so the findings of a meta-analysis of LPD in persons with diabetes are important. In this study, Pan and colleagues⁷³ assessed eight RCTs. In contrast to a previous meta-analysis, they found that in comparison with a normal-protein diet,

treatment with an LPD was not associated with a significant improvement in kidney function (assessed by GFR). However, in only two of these trials, there was a significant but marginal decrease in proteinuria or albuminuria in the participants following the LPD. Of importance was that persons following the LPD had lower serum albumin levels and poorer glycemic control, both of which are relevant to outcomes and their assessment.

Fouque and Laville⁷⁴ performed a Cochrane Database systematic review of all randomized studies, comparing two different levels of protein intake in adult persons suffering from moderate to severe CKD. A total of 2000 nondiabetic persons were identified in 10 studies (of a total of 40 studies) in which follow-up lasted at least 1 year. There were 281 renal deaths (progression to ESKD) recorded, 113 among participants following the LPD and 168 among those following the higher protein diet (RR, 0.68; 95% CI, 0.55–0.84; $P = .0002$). The authors concluded that reducing protein intake in persons with CKD reduces the occurrence of renal death by 32% in comparison with higher or unrestricted protein intake. The most recent meta-analysis included 16 RCTs of persons with relatively advanced CKD (stage 4 and 5 in most cases) not receiving RRT. Inclusion required at least 30 participants per study. In studies comparing LPD (<0.8 g/day) with higher protein intake, LPD was associated with a small but significant absolute risk reduction for ESKD (4%) and higher serum bicarbonate at 1 year (weighted mean difference [MD], 1.46 mEq/L) versus higher protein diets. Similarly, in studies comparing VLPD with LPD, VLPD was associated with an absolute risk reduction for ESKD (13%) and higher GFR at 1 year (weighted MD, 3.95 mL/min/1.73 m²) versus LPD. No studies reported increased risk of protein energy wasting or other safety concerns associated with dietary protein restriction.⁷⁵

The caveats for these trials are that most were of short duration, the largest trial (MDRD) mostly excluded persons with diabetes, and compliance with LPDs was a factor in interpretation in some of the results. Furthermore, the proportion of the participants treated with renin–angiotensin–aldosterone system inhibitors (RAASi) was variable. Available evidence suggests that LPD and RAASi may have synergistic renoprotective effects but this has not yet been adequately evaluated in an RCT.⁷⁶ In addition to possible renoprotective effects, dietary protein restriction results in reduced intake of sodium, phosphate, and acid all of which may be beneficial in CKD.⁷⁷

In summary, published data indicate that there is likely some renoprotective benefit from dietary protein restriction but conclusive evidence is lacking, in particular for additional benefit in persons already treated with RAASi. The benefits are more apparent in persons with proteinuria and lower GFR, and the risks are greater in those at risk of malnutrition, such as elderly persons (age >75 years); those with below-average BMI (<20 kg/m²), muscle wasting, or myopathic symptoms; and those with evidence of protein-energy wasting. We therefore recommend that the potential benefits and risks should be considered carefully on an individual basis and with the assistance of a trained renal dietitian. The 2012 KDIGO guideline recommends reducing dietary protein intake to less than 0.8 g/kg/day in adults with stages 4 and 5 CKD, and avoiding a high-protein diet (>1.3 g/kg/day) in adults with CKD who are considered to be at risk of

progression.⁵ By contrast, the 2014 NICE guideline recommends against dietary protein restriction for all persons with CKD.⁷⁸ For further discussion of dietary aspects in kidney disease, see Chapter 60.

GLYCEMIC CONTROL IN PERSONS WITH DIABETES

The role of glycemic control in protecting the kidneys in persons with diabetes is discussed fully in Chapter 39. In summary, the benefit of tight glycemic control in ameliorating diabetic nephropathy seems to decrease as CKD progresses; the greatest benefits are observed in stages 1 and 2. In addition to the renoprotective effects, however, there is clear evidence that improving glycemic control can reduce the risk of developing other microvascular and macrovascular complications such as blindness and cardiovascular disease.⁷⁹ Of importance is that these benefits were maintained for up to 10 years after the trial, even though intensive control regressed to standard control levels.⁸⁰ Achieving optimal glycemic control should therefore be an important goal for all persons with diabetes and CKD, but should be balanced against the risk of developing hypoglycemia, which may be increased in the elderly and those with more advanced CKD. The KDIGO guidelines recommend a target hemoglobin A_{1c} of 7.0% (53 mmol/mol) to prevent or delay progression of microvascular complications, including diabetic kidney disease, unless persons have multiple comorbidities, have reduced life expectancy, or are at risk of hypoglycemia.⁵

ANTIHYPERTENSIVE THERAPY

The treatment of systemic hypertension was the first intervention shown to significantly slow the rate of CKD progression and remains fundamental in renoprotective strategies. Mogensen⁸¹ and Parving and colleagues⁸² pioneered the role of blood pressure control in studies of persons with type 1 diabetes, in whom the initiation of antihypertensive therapy significantly slowed the rate of GFR decline.

Similar observations were subsequently reported among persons with nondiabetic forms of CKD.^{83–85} In one of the earliest meta-analyses in nephrology, Kasiske and colleagues⁸⁶ studied 100 controlled and uncontrolled studies that provided data on kidney function, proteinuria, or both before and after treatment with an antihypertensive agent in people with diabetes. Multiple linear regression analysis indicated that ACE inhibitors decreased proteinuria independently of changes in blood pressure, treatment duration, type of diabetes or stage of nephropathy, and study design. Reductions in proteinuria from other antihypertensive agents could be attributed entirely to changes in blood pressure. Furthermore, blood pressure reduction was associated with a relative increase in GFR (3.70 ± 0.92 mL/min for each 10-mm Hg reduction in mean arterial pressure [MAP]; $P = .0002$), but in comparison with other agents, ACE inhibitors had an additional favorable effect on GFR that was independent of blood pressure changes (3.41 ± 1.71 mL/min; $P = .05$).

ANTIHYPERTENSIVE DRUGS

Data from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT) have been misinterpreted by some as implying that the choice of antihypertensive drug does not affect renal outcomes in persons with

CKD. Note that ALLHAT was designed to investigate the effect of antihypertensive drugs on cardiovascular rather than renal outcomes in persons who had hypertension and at least one cardiovascular risk factor. There was no significant difference in the incidence of the primary outcome of fatal or nonfatal myocardial infarction among persons randomly assigned to receive treatment with a thiazide diuretic, a calcium channel blocker, or an ACE inhibitor.⁸⁷ Whereas a post hoc analysis also showed no difference in the secondary outcome of ESKD or more than a 50% decrease in GFR, persons with serum creatinine levels exceeding 2 mg/dL (>170 μ mol/L) were specifically excluded from the study, and therefore only a minority (5662 of 33,357) had CKD, mostly in stages 1 to 3. Furthermore, the presence of proteinuria was not assessed.⁸⁸ By contrast, a large body of evidence supports the use of ACE inhibitors or ARB agents as first-line antihypertensive therapy in persons with CKD (see the “[Pharmacologic Inhibition of the Renin–Angiotensin–Aldosterone System](#)” section under “[Interventions for Slowing Progression of Chronic Kidney Disease](#)”).

Despite the importance of ACE inhibition and ARB therapy in achieving renoprotection, thiazide and other diuretics are valuable and sometimes essential as additional antihypertensive agents to achieve optimal blood pressure control. Studies have shown that high dietary sodium intake may abrogate the antiproteinuric effect of ACE inhibitor treatment, but addition of a thiazide diuretic restores the antiproteinuric effect despite ongoing high sodium intake.⁵⁷ Similarly, addition of a thiazide diuretic to ARB treatment reduced blood pressure and proteinuria in persons with IgA nephropathy.⁸⁹ We therefore recommend a thiazide diuretic as second-line antihypertensive therapy in persons who have not achieved adequate blood pressure control with an ACE inhibitor or an ARB alone. In persons with advanced CKD, thiazide diuretics are considered less likely to be effective, although a recent review of published literature suggests that they may still be beneficial in stage 4 CKD.⁹⁰ In those who do not respond to a thiazide diuretic, a loop diuretic should be considered instead.

There is some evidence that, despite their efficacy as antihypertensive drugs, dihydropyridine calcium channel blockers (DCCBs) may have adverse effects with regard to CKD progression. In experimental studies, DCCB treatment was observed to allow greater transmission of systemic blood pressure to glomerular capillaries and was associated with more rapid progression of renal injury than was ACE inhibitor treatment in the 5/6 nephrectomy model.⁹¹ Whereas one relatively small study revealed no difference between the renoprotective effects of the DCCB nifedipine and the ACE inhibitor captopril,⁹² two larger studies demonstrated adverse outcomes associated with the use of DCCB agents.

A secondary analysis of data from the REIN study revealed that among participants not randomly assigned to receive ACE inhibitor treatment who failed to achieve a MAP of less than 100 mm Hg, the DCCB agents nifedipine and amlodipine were associated with a higher magnitude of proteinuria and more rapid GFR decline than were other antihypertensives.⁹³ Observations from the African American Study of Kidney Disease and Hypertension (AASK) provoked even greater concern.⁹⁴ In this study, persons with CKD and hypertension were randomly assigned to receive either treatment with an ACE inhibitor or amlodipine (DCCB) or treatment with a

β -blocker and diuretic in combination. The amlodipine arm of the study was stopped prematurely because recipients exhibited a more rapid decline in GFR than did participants receiving the β -blocker or ACE inhibitor, particularly among those with urine protein levels higher than 1 g/day. The Ramipril Efficacy In Nephropathy-2 (REIN-2) study revealed no additional renoprotection when a DCCB was added to ACE inhibitor treatment in persons with nondiabetic CKD, but there was also no adverse effect.⁹⁵ By contrast, treatment with nondihydropyridine calcium channel blockers (NDCCBs) ameliorated glomerular hypertension, reduced proteinuria, and afforded renoprotection in some experimental studies.⁹⁶ In one clinical study, the combination of ACE inhibition and NDCCB treatment resulted in greater reduction of proteinuria than did either of these treatments alone in persons with type 2 diabetes and overt nephropathy.⁹⁷ Furthermore, a meta-analysis of data from 28 RCTs revealed a 2% increase in proteinuria with DCCB treatment, in contrast to a 30% reduction with NDCCB treatment, despite similar effects on blood pressure in hypertensive persons with proteinuria.⁹⁸

On the basis of the evidence discussed, we recommend that DCCBs be avoided in persons with CKD unless they are used in combination with ACE inhibition or ARB treatment to achieve adequate blood pressure control. If possible, NDCCBs should be used in preference over DCCBs. Most of the trials in persons with CKD have established that between two and four agents is the median number of antihypertensive agents required for the respective target blood pressures. Therefore third-line agents are needed by a substantial proportion of persons with CKD. We recommend that the choice of third- and fourth-line agents be based on factors other than renoprotection. Therefore in addition to RAASi, diuretic therapy, and NDCCBs, we suggest use of β -blockers, α -blockers, and central nervous system agents, depending on comorbid conditions, side-effect profile, drug interactions (in particular, NDCCBs should not be used in combination with β -blockers due to increased risk of cardiac conduction abnormalities), convenience, and cost.

TRIALS OF LOW VERSUS USUAL TARGETS FOR LOWERING BLOOD PRESSURE

Despite the importance of blood pressure control in achieving renoprotection, the optimal blood pressure remains uncertain, particularly in elderly persons and those with mild proteinuria. In several RCTs, researchers have investigated whether blood pressure targets lower than previously recommended levels afford greater renoprotection than “usual” blood pressure control, but these studies have not provided a conclusive answer. In the MDRD study, primary analysis showed no significant difference in the rate of GFR decline between participants aged 18 to 60 years randomly assigned to achieve a target MAP of less than 92 mm Hg (equivalent to <125/75 mm Hg), or MAP less than 98 mm Hg or participants aged 61 years or older assigned a MAP target of less than 107 mm Hg (equivalent to 140/90 mm Hg), or MAP less than 113 mm Hg. Participants assigned to achieve the lower blood pressures, however, evidenced an early rapid decrease in GFR, probably as a result of associated renal hemodynamic effects, that obscured a later slower rate of GFR decline. Furthermore, secondary analysis did show benefit associated with the lower blood pressure target among persons with more severe baseline proteinuria (urine protein level >1 g/

day). Further secondary analysis revealed that lower achieved blood pressure was also associated with a slower GFR decline, an effect that was more marked among persons with more severe baseline proteinuria.⁶⁷ The authors concluded by recommending a blood pressure goal of less than 125/75 mm Hg (MAP, 92 mm Hg) for persons with CKD whose urine protein levels exceed 1 g/day, and a goal of less than 130/80 mm Hg (MAP, 98 mm Hg) for those with urine protein levels of 0.25 to 1.0 g/day.⁹⁹

Findings of prolonged follow-up of the participants in the MDRD study suggest that the benefits of lower blood pressure may become evident only over a longer period. Analysis after almost 10 years revealed a significant reduction in the risk of ESKD (adjusted HR, 0.68) and a combined endpoint of ESKD or death (adjusted HR, 0.77) among persons randomly assigned to achieve the lower blood pressure targets. An important caveat is that treatment and blood pressure data were not available beyond the 2.2 years of the original trial.¹⁰⁰ In the AASK, no significant difference in the rate of GFR decline was observed among participants randomly assigned to achieve a MAP goal of 92 mm Hg or lower versus a goal of 102 to 107 mm Hg. One possible explanation for this outcome is that participants in the AASK generally had milder baseline proteinuria (mean urine protein excretion, 0.38–0.63 g/day).¹⁰¹ The results are therefore consistent with those of the MDRD study, which showed benefit only in persons with significant proteinuria. Likewise, in the REIN-2 trial, additional blood pressure reduction (blood pressure of <130/80 mm Hg vs. DBP of <90 mm Hg, regardless of SBP) failed to provide additional renoprotection in persons with nondiabetic CKD who were already receiving ACE inhibitor treatment.⁹⁵ Possible explanations are that the degree of additional blood pressure reduction was modest (4.1/2.8 mm Hg) and that in the group undergoing intensive blood pressure reduction, the number of participants with moderate to heavy proteinuria was small.

The Systolic Blood Pressure Intervention Trial (SPRINT) is the largest randomized trial to date to compare the effect of different BP targets on cardiovascular and renal outcomes in persons with CKD. The study included 9361 participants aged 50 years and over with BP 130 to 180 mm Hg and increased cardiovascular risk (including all with CKD) who were randomly allocated to a systolic BP target of less than 120 mm Hg or less than 140 mm Hg. Persons with diabetes, proteinuria of greater than 1 g/day, adult polycystic kidney disease, previous stroke, and cardiac failure were excluded. The main trial was stopped early due to evidence of significant benefit in the primary outcome of CVE and all-cause mortality in the lower BP group.¹⁰² A prespecified subgroup analysis was conducted in 2646 participants with CKD at baseline. Analysis of the main trial showed no evidence of effect modification by CKD status on the primary outcome of CVE or all-cause mortality. In addition, in the CKD subgroup, participants in the lower BP group evidenced a lower incidence of CVE (HR, 0.81; 95% CI, 0.63–1.05) and all-cause mortality (HR, 0.72; 95% CI, 0.53–0.99; Fig. 59.4). Importantly, these benefits were also observed in participants aged 75 years and older. There was no difference between groups in the renal outcome of 50% decline in GFR or ESKD after a median of 3.3 years. The lower BP target was associated with some initial decline in eGFR, likely due to increased use of RAASi, whereas a small initial increase in GFR was observed with the higher

BP. After excluding the first 6 months of observation the low BP target was associated with a slightly higher rate of GFR decline (0.47 mL/min/year vs. 0.32 mL/min/year; $P < .03$). The lower BP target was associated with lower UACR at all time points to 48 months. There was no difference in serious adverse events between the groups but the low BP target was associated with a higher incidence of hyperkalemia, hypokalemia, and AKI¹⁰³ (predominantly stage 1 and 2 and most had recovery of kidney function).¹⁰⁴ In summary, this largest randomized trial to date found that a lower BP target than previously recommended was associated with a significant reduction in CVE and all-cause mortality. There was no benefit in renal outcomes but also no evidence of harm (apart from a small increase in incidence of AKI). It should be remembered that persons with advanced or severe CKD (evidenced by proteinuria >1 g/day) as well as those with diabetes were excluded. The findings are therefore consistent with those of the MDRD and AASK studies, confirming that there is little renoprotective benefit associated with low BP targets in mild and nonproteinuric CKD. However, the results do indicate significant survival and cardiovascular benefit with lower BP targets and question the notion that the risk of lower BP targets outweighs the benefits in all older people.

Because not all the persons in the aforementioned trials received ACE inhibitor treatment, it remained unclear how important the level of blood pressure attained was in persons with CKD who were receiving an ACE inhibitor or an ARB. Several studies have sought to address this issue. Among persons with type 1 diabetes and established nephropathy who were receiving ACE inhibitor treatment, attainment of a “low” (MAP, 92 mm Hg) versus “usual” (MAP, 100–107 mm Hg) target blood pressure was associated with significantly milder degrees of proteinuria after 2 years, but there was no significant difference in GFR.¹⁰⁵ The Effect of Strict Blood Pressure Control and ACE Inhibition on the Progression of Chronic Renal Failure in Pediatric Patients (ESCAPE) trial was conducted to investigate the role of blood pressure control in children with CKD who were receiving treatment with an ACE inhibitor. Attainment of a lower blood pressure target was associated with a significantly reduced risk of reaching ESKD or doubling of serum creatinine concentration.¹⁰⁶ This is an important trial inasmuch as the primary endpoint (time to 50% decline in GFR or progression to ESKD) was not as affected by the competing mortality effects that complicate studies in older persons. A total of 30% of the participants who received intensified treatment for blood pressure control reached the primary endpoint, in comparison with 42% of those who received conventional treatment for blood pressure control (HR, 0.65; 95% CI, 0.44–0.94; $P = .02$). Urine protein excretion gradually rebounded during ongoing ACE inhibition after an initial 50% decrease, despite persistently good blood pressure control. Achievement of blood pressure targets and a decrease in proteinuria were significant independent predictors of delayed progression of CKD.

Results of secondary analyses of other studies also indicate that lower blood pressure is associated with more effective renoprotection in persons receiving ACE inhibitors or ARB treatment. In the Irbesartan in Diabetic Nephropathy Trial (IDNT), greater renoprotection was observed in persons who achieved lower blood pressure: Achieved SBP greater than 149 mm Hg was associated with a 2.2-fold increased risk of

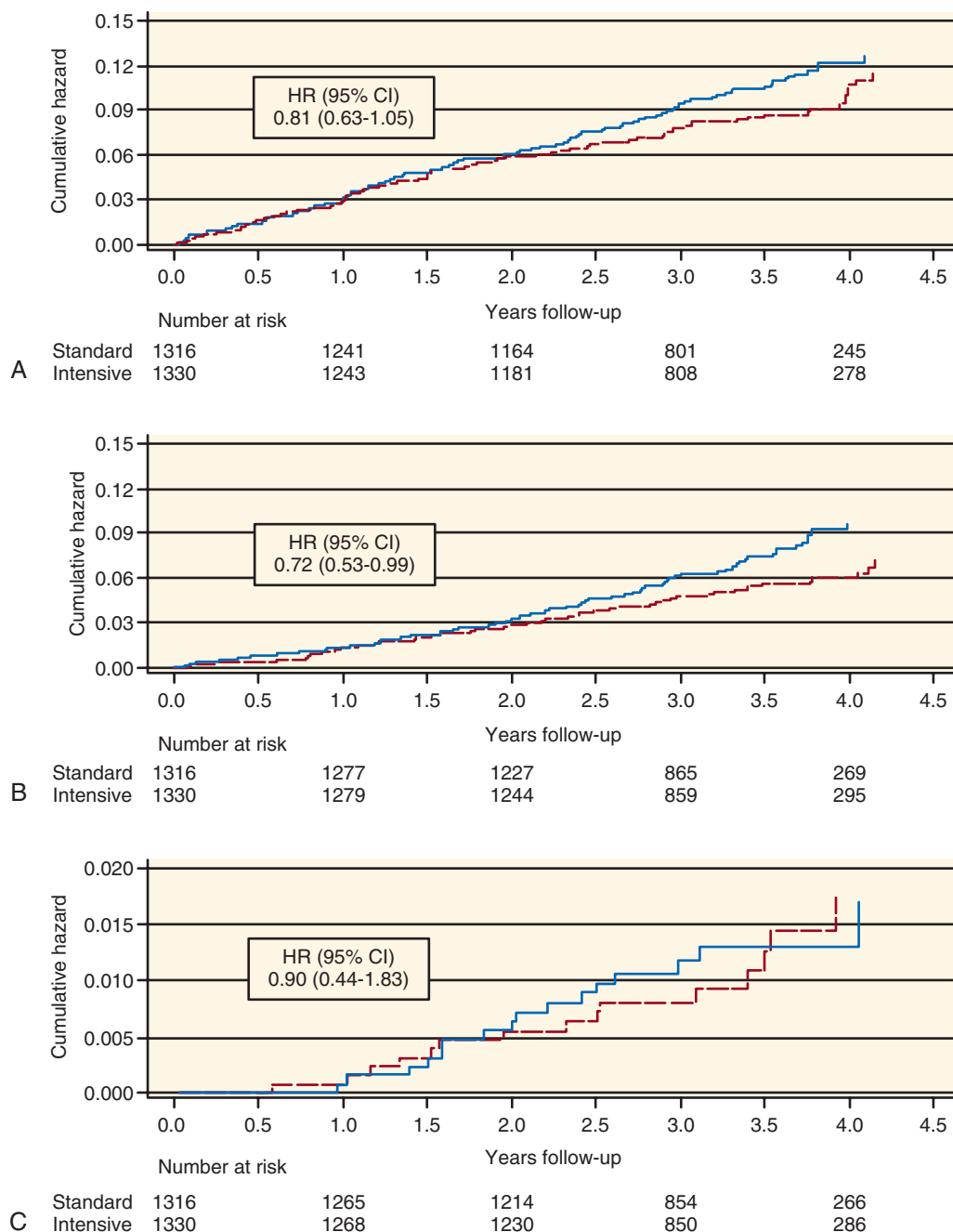


Fig. 59.4 Kaplan-Meier curves for prespecified outcomes in the Systolic Blood Pressure Intervention Trial (SPRINT) participants with chronic kidney disease. (A) The primary cardiovascular outcome, defined as the composite of myocardial infarction, acute coronary syndrome, stroke, acute decompensated heart failure, and death from cardiovascular causes. (B) The all-cause death outcome. (C) The main kidney outcome, defined as the composite of a decrease in estimated glomerular filtration rate of 50% or more from baseline (confirmed by repeat testing ≥ 90 days later) or the development of end-stage kidney disease. The broken lines depict the intensive blood pressure treatment group [systolic blood pressure (SBP) less than 120 mm Hg]; the solid lines depict the standard blood pressure treatment group (SBP < 140 mm Hg). CI, Confidence interval; HR, hazard ratio. (From Cheung AK, Rahman M, Reboussin DM, et al. Effects of intensive BP control in CKD. *J Am Soc Nephrol.* 2017;28:2812-2823.)

developing ESKD or of serum creatinine doubling, in comparison with an achieved SBP less than 134 mm Hg, independent of ARB treatment.¹⁰⁷ Progressive lowering of SBP to 120 mm Hg was associated with improved renal outcome and improved rates of patient survival, an effect independent of baseline GFR. However, the lower blood pressure observed was not a primary aim of these studies, and it cannot be

assumed that this observed association (between lower blood pressure and improved renal and patient outcomes) is causative.

Several meta-analyses have combined data from multiple trials to examine the potential benefit of lower BP targets in persons with CKD. In one study that included 11 randomized trials with 1860 participants who had nondiabetic CKD

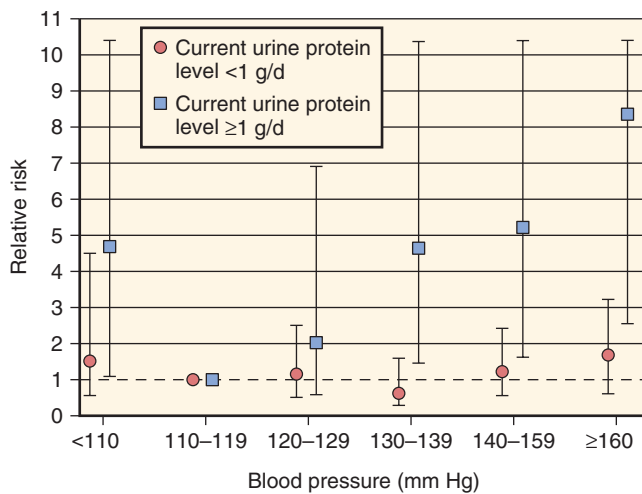


Fig. 59.5 Results of a meta-analysis of 11 randomized trials that included 1860 persons with nondiabetic chronic kidney disease, showing the relationship between relative risk for kidney disease progression (doubling of serum creatinine level or end-stage kidney disease) and achieved systolic blood pressure in two groups according to the magnitude of proteinuria. (From Jafar TH, Stark PC, Schmid CH, et al. Progression of chronic kidney disease: the role of blood pressure control, proteinuria, and angiotensin-converting enzyme inhibition: a patient-level meta-analysis. *Ann Intern Med.* 2003;139:244–252.)

and proteinuria greater than 1 g/day, the lowest risk of progression was associated with an achieved SBP of 110 to 129 mm Hg, independent of ACE inhibitor treatment (Fig. 59.5).¹⁰⁸ A more recent analysis that included 8127 participants from nine trials (including SPRINT) did not show a significant difference between intensive and usual BP control on the annual rate of change in GFR (MD, 0.07; 95% CI, −0.16 to 0.29 mL/min/year), doubling of serum creatinine concentration or 50% reduction in GFR (RR, 0.99; 95% CI, 0.76–1.29), ESKD (RR, 0.96; 95% CI, 0.78–1.18), composite renal outcome (RR, 0.99; 95% CI, 0.81–1.21), or all-cause mortality (RR, 0.81; 95% CI, 0.64–1.02). Intensive BP control did reduce mortality (RR, 0.78; 95% CI, 0.61–0.99) in a sensitivity analysis that excluded participants with diabetes. In addition, a trend toward a slower rate of GFR decline was observed in those with proteinuria greater than 1 g/day.¹⁰⁹ By contrast, another recent meta-analysis of 18 randomized trials involving 15,924 participants with and without diabetes reported a 14.0% lower risk of all-cause mortality associated with intensive BP lowering (odds ratio [OR], 0.86; 95% CI, 0.76–0.97; $P = .01$). These findings remained similar when data were reanalyzed without the data from SPRINT.¹¹⁰ The difference in results between meta-analyses is likely attributable to the inclusion of different trials in each, but they are broadly in keeping with the findings of the individual trials discussed earlier.

By contrast, several sources indicate that excessive lowering of blood pressure may be associated with adverse effects in some persons with CKD. In one meta-analysis,¹⁰⁸ an achieved SBP lower than 110 mm Hg was associated with an increased risk of CKD progression (RR, 2.48; 95% CI, 1.07–5.77; see Fig. 59.5), and in IDNT, an achieved SBP lower than 120 mm Hg was associated with increased rates of all-cause mortality and no further improvement in renal outcomes.¹⁰⁷ In addition, secondary analysis of data from the Ongoing Telmisartan

Alone and in Combination with Ramipril Global Endpoint Trial (ONTARGET) revealed that hypertensive persons with risk factors for cardiovascular disease who achieved an SBP of less than 120 mm Hg had a significantly higher rate of cardiovascular mortality than did those who achieved an SBP of 120 to 129 mm Hg.¹¹¹ Similarly, the Action to Control Cardiovascular Risk in Diabetes (ACCORD) blood pressure study reported no difference in primary cardiovascular outcomes (nonfatal myocardial infarction, nonfatal cerebrovascular accident, or cardiovascular death) among persons with diabetes randomly assigned to achieve an SBP target of less than 120 mm Hg, in comparison with a conventional control target of less than 140/80 mm Hg. During follow-up, the mean SBP was 119.3 mm Hg in persons undergoing intensive therapy and 133.5 mm Hg in those undergoing standard therapy. Persons undergoing intensive therapy demonstrated a slightly lower stroke rate (annual rates of 0.32% vs. 0.53%) but a higher rate of treatment-related adverse events (3.3% vs. 1.3%). Specifically, there were significantly more instances of an eGFR of less than 30 mL/min/1.73 m² among persons receiving intensive therapy than among those receiving standard therapy (99 vs. 52 events; $P < .001$) and no difference in the primary cardiovascular composite outcome.¹¹²

The randomized trials in which “low” and “usual” blood pressure targets were compared among persons with CKD have not yielded unequivocal results, but overall, lower blood pressure targets were associated with more effective renoprotection, particularly among persons with significant proteinuria, and lower risk of CVE and/or all-cause mortality. By contrast, there is also an increase in adverse events in some but not all studies. Starting with the observation that persons most likely to benefit from lower BP targets are likely different from those most at risk of adverse events, Basu et al. developed a model for predicting those who would most likely benefit from lower BP targets using data from SPRINT and validated this using data from ACCORD. Older age, black race, higher diastolic BP, and higher lipids were associated with greater cardiovascular risk reduction and current smoking was associated with less cardiovascular risk reduction (C-statistic, 0.71; 95% CI, 0.68–0.74), whereas male sex, current smoking, statin use, elevated creatinine, and higher lipids were associated with higher risk of adverse events (C-statistic, 0.71; 95% CI, 0.69–0.73). In the SPRINT cohort, 4.7% of participants were in the highest two benefit subgroups but the lowest subgroup of harm and 25.7% were in the lowest benefit subgroup (no significant benefit) but the highest two harm subgroups. By contrast, in the ACCORD cohort only 1.5% of participants were in the highest benefit subgroup but the lowest subgroup of harm and 60.9% were in the two lowest benefit subgroups (no significant benefit) but the highest two harm subgroups.¹¹³ An online risk assessment tool has been developed to facilitate the assessment of benefit versus risk.¹¹⁴

Current guidelines differ somewhat in their recommendations for blood pressure targets. The KDIGO guidelines (which predate the publication of SPRINT) recommend an SBP of 140 mm Hg or less and DBP of 90 mm Hg or less for diabetic and nondiabetic adults with CKD and urine albumin excretion of less than 30 mg/day (or equivalent). For adults with diabetic or nondiabetic CKD and albumin excretion of at least 30 mg/day, the target values are 130 mm Hg or lower and 80 mm

Hg or lower, respectively.⁵ The 2017 American College of Cardiology/American Heart Association guidelines did not adopt the low systolic BP target of less than 120 mm Hg supported by SPRINT but did recommend a target of less than 130/80 mm Hg for persons with increased cardiovascular risk, including those with diabetes and/or CKD.¹¹⁵ We support these recommendations with a caveat that treatment should be individualized and that low BP targets should be avoided in those at increased risk of adverse effects.

Clinical Relevance

Evidence from the Systolic Blood Pressure Intervention Trial (SPRINT) indicates that a lower target for the treatment of systolic blood pressure may afford improved cardiovascular risk reduction but may also be associated with some increase in risk of acute kidney injury and hyperkalemia. The risks versus benefits of a lower blood pressure target should be evaluated in each individual.

PHARMACOLOGIC INHIBITION OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

A large number of published clinical trials and meta-analyses provide clear evidence to support the use of pharmacologic inhibitors of the RAAS as an essential component of any

strategy aiming to achieve maximal renoprotection in persons with CKD (Table 59.1).

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS

Diabetic Kidney Disease

In 1993 the Captopril Collaborative Study Group published results of the first large prospective RCT to clearly show specific renoprotection attributable to ACE inhibitor treatment, a landmark event in the development strategies for achieving renoprotection in persons with diabetes and CKD.¹¹⁶ Persons ($n = 409$) with type 1 diabetes and established nephropathy (urine protein excretion >0.5 g/day; serum creatinine levels <2.5 mg/dL) were randomly assigned to receive captopril or placebo, and a blood pressure goal of less than 140/90 mm Hg was set for both groups. After a median follow-up period of 3 years, captopril treatment was associated with a 50% reduction in the risk of the combined endpoint of death, dialysis, and renal transplantation and a 48% reduction in the risk of serum creatinine doubling (Fig. 59.6). Because blood pressure control was not statistically different between the groups, the additional renoprotection was not attributable simply to the antihypertensive effects of ACE inhibitors.

These results prompted several further studies to investigate whether ACE inhibitors may also benefit persons with early stage nephropathy characterized by microalbuminuria. A meta-analysis of 12 such studies, including 689 persons with

Table 59.1 Summary of Studies Showing the Renoprotective Effects of Angiotensin-Converting Enzyme Inhibitors and Angiotensin Receptor Blockers in Diabetic and Nondiabetic Persons With Chronic Kidney Disease

CKD Type	Trial Outcome	Reference
Angiotensin-Converting Enzyme Inhibitors		
Type 1 DM + CKD	↓ Risk of dialysis or death	Lewis et al. ¹¹⁶
Type 1 DM + microalbuminuria	↓ Risk of overt nephropathy	Mathiesen et al. ^{116a} Laffel et al. ^{116b} Viberti et al. ^{116c}
Type 1 DM + normoalbuminuria	No significant benefit	Mauer et al. ¹³² EUCLID Study Group ^{116d}
Type 2 DM + CKD	Benefit in 1 study only	Bakris et al. ¹¹⁸
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Ravid et al. ^{125,126} Sano et al. ¹²² Trevisan and Tiengo ¹²² Agardh et al. ¹²⁴ Ahmad et al. ¹²⁷ Ruggenenti ¹³⁰
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria	Ravid et al. ¹²⁸
Nondiabetic CKD	↓ Doubling of creatinine level/ESKD	Ruggenenti et al. ^{138,139}
Angiotensin Receptor Blockers		
Type 1 DM + normoalbuminuria	Small ↑ in albuminuria or no benefit	Mauer et al. ¹³² Bilous et al. ¹⁴⁷
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria (small ↑ in rate of cardiovascular mortality)	Bilous et al. ¹⁴⁷ Haller et al. ¹⁴⁸
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Parving et al. ¹⁴⁶
Type 2 DM + CKD	↓ Risk of doubling of creatinine level ↓ Risk of ESKD	Lewis et al. ¹⁴⁵ Brenner et al. ¹⁴⁴

CKD, Chronic kidney disease; DM, diabetes mellitus; ESKD, end-stage kidney disease.

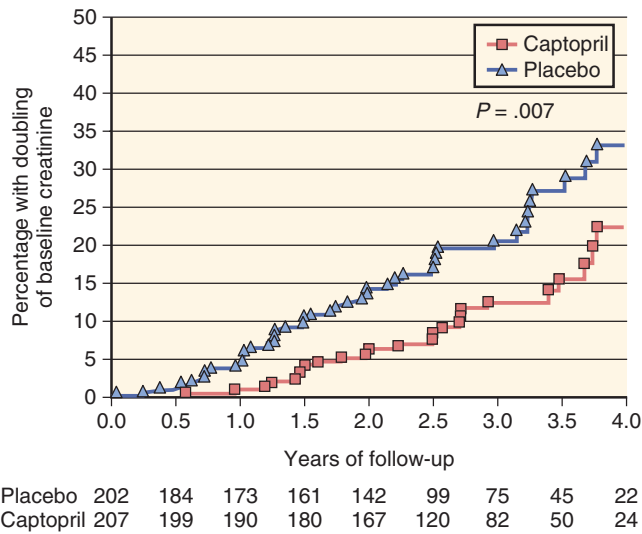


Fig. 59.6 Results of the first randomized controlled trial to show the renoprotective effects of angiotensin-converting enzyme inhibitor treatment, independent of blood pressure lowering. Persons with type 1 diabetes and nephropathy were randomly assigned to receive treatment with captopril or placebo, and blood pressure was matched between the groups. The graph shows that the cumulative incidence of the primary endpoint, doubling of serum creatinine level, was significantly lower in the persons who received captopril treatment. (From Lewis EJ, Hunsicker LG, Bain RP, et al. The effect of angiotensin-converting-enzyme inhibition on diabetic nephropathy. *N Engl J Med.* 1993;329:1456–1462.)

type 1 diabetes who were monitored for at least 1 year, revealed that ACE inhibitor treatment was associated with a significant reduction in the risk of progression to overt nephropathy (OR, 0.38) and three times the incidence of normalization of microalbuminuria.¹¹⁷

Data regarding the renoprotective effects of ACE inhibitors in persons with type 2 diabetes are, to some extent, equivocal. Comparisons of ACE inhibitors and other antihypertensives among persons with overt nephropathy have included relatively small numbers of participants, and only one¹¹⁸ demonstrated greater reduction in GFR decline in association with ACE inhibitor treatment.^{119–121} Evidence of renoprotective benefit at earlier stages of nephropathy is more consistently supportive. Several studies, including the diabetic subgroup analysis of the Heart Outcomes Prevention Evaluation (HOPE) study, demonstrated beneficial effects of ACE inhibitor treatment among persons with type 2 diabetes in decreasing microalbuminuria^{122–124} or in reducing the number of persons progressing from microalbuminuria to overt proteinuria (risk reduction, 24% to 67%).^{125–128} In addition, the HOPE study reported a 25% reduction in the combined primary endpoint of myocardial infarction, stroke, or cardiovascular death in ramipril-treated persons with type 2 diabetes and risk factors for cardiovascular disease.

Two studies revealed a beneficial role for ACE inhibitor treatment in primary prevention of nephropathy in persons with type 2 diabetes. Among 156 normotensive, normoalbuminuric persons, ACE inhibitor treatment was associated with a 12.5% absolute risk reduction for microalbuminuria,^{128,129} and in 1204 hypertensive normoalbuminuric persons, the addition of ACE inhibitor treatment to verapamil was associ-

ated with a lower incidence of microalbuminuria than was placebo plus verapamil.¹³⁰ By contrast, one relatively large study demonstrated no renoprotective benefit of ACE inhibitors over β -blocker treatment among hypertensive persons with type 2 diabetes with normoalbuminuria or microalbuminuria,¹³¹ and another revealed no reduction in the incidence of microalbuminuria in persons with type 1 diabetes randomly assigned to receive treatment with an ACE inhibitor versus placebo.¹³² Three meta-analyses were conducted to investigate the effect of ACE inhibitor treatment in persons with diabetic nephropathy. One analysis included only studies of type 2 diabetes with albuminuria or proteinuria as an outcome; it revealed statistically significant reductions in albuminuria in association with ACE inhibitor treatment versus placebo.¹³³ A second, larger analysis combined data from studies of type 1 and type 2 diabetes and revealed weak evidence of a reduced risk of serum creatinine doubling (RR, 0.60; 95% CI, 0.34–1.05) or reduced incidence of ESKD (RR, 0.64; 95% CI, 0.40–1.03) and stronger evidence of reduced risk of progression of microalbuminuria to macroalbuminuria (RR, 0.45; 95% CI, 0.28–0.71) with ACE inhibitor treatment versus placebo. All-cause mortality was significantly reduced in persons receiving ACE inhibitors (RR, 0.79; 95% CI, 0.63–0.99).¹³⁴ A third meta-analysis involved data from 16 studies of the effect of ACE inhibitor treatment on reducing the risk of microalbuminuria in type 1 and type 2 diabetes. This meta-analysis revealed a significantly reduced risk of developing microalbuminuria with ACE inhibitors in comparison with placebo (RR, 0.60; 95% CI, 0.43–0.84) or calcium channel blocker treatment (RR, 0.58; 95% CI, 0.40–0.84).¹³⁵

On the basis of the data just described, we recommend ACE inhibitor treatment as first-line therapy for all persons with type 1 diabetes and microalbuminuria or overt nephropathy. At present, data are insufficient to support the use of ACE inhibitors to prevent nephropathy in normoalbuminuric persons with type 1 diabetes, but it seems reasonable to recommend them as the treatment of choice in those with hypertension. There is, however, sufficient evidence to recommend the use of ACE inhibitors to reduce progression to overt nephropathy in persons with type 2 diabetes and microalbuminuria or to prevent microalbuminuria in those with hypertension. There is no clear evidence of specific benefit associated with ACE inhibitors in slowing the progression of overt nephropathy in persons with type 2 diabetes; this lack of evidence may be ascribable to the lack of adequately powered studies. Finally, because cardiovascular disease is the most common cause of morbidity and mortality among persons with type 2 diabetes, ACE inhibitor treatment should be considered for the reduction of cardiovascular risk. The KDIGO guidelines recommend treatment with an ACE inhibitor or ARB in all adults with diabetes and urine albumin excretion 30 mg/day or more (or equivalent).⁵ For further discussion of the management of diabetic nephropathy, see Chapter 39.

Nondiabetic Chronic Kidney Disease

After reports of renoprotection with ACE inhibitor treatment in diabetic nephropathy, further studies sought to investigate the renoprotective potential of ACE inhibitors in nondiabetic forms of CKD. One early study demonstrated a 53% reduction in risk of the composite endpoint (serum creatinine doubling or ESKD) in association with ACE inhibitor treatment, but

a significantly lower blood pressure in persons receiving ACE inhibitors versus placebo made it impossible to separate the beneficial effects of lowering blood pressure from any unique effects of ACE inhibitor treatment.¹³⁶ By contrast, in the REIN study of 352 persons with nondiabetic CKD and urine protein levels exceeding 1 g/day, similar control of blood pressure was achieved in the participants randomly assigned to receive an ACE inhibitor or placebo. Among persons with at least 3 g/day of urine protein at baseline, the study was stopped early because the rate of decline in GFR in participants receiving the ACE inhibitor was significantly lower (0.53 vs. 0.88 mL/min/month)¹³⁷; further analysis showed a significantly lower risk of the combined endpoint (serum creatinine doubling or ESKD) in the participants taking ACE inhibitors (RR, 1.91; 95% CI, 1.10–3.33) for the placebo recipients; Fig. 59.7).

One hundred eighty-six persons from the REIN study who had less than 3 g/day of urine protein were monitored for a median of 31 months after randomization. In findings similar to those of participants with more severe proteinuria, ACE inhibitor treatment significantly reduced the incidence of ESKD (for placebo recipients, RR, 2.72; 95% CI, 1.22–6.08), particularly among those with a GFR of less than 45 mL/min at baseline.¹³⁸ After the randomized phase of the study, persons who had received placebo were switched to ACE inhibitors, and those taking ACE inhibitors continued treatment. In a finding consistent with those of the first phase of the study,

there was a significant reduction in the rate of decline in GFR of persons switched to ACE inhibitors. In addition, persons continuing with ACE inhibitor treatment showed a further reduction in the rate of GFR decline. Persons who had received ACE inhibitors from the start of the REIN study had a significantly lower risk of reaching ESKD than did those who switched to ACE inhibitors after the initial phase (for placebo recipients, RR, 1.86; 95% CI, 1.07–3.26). In fact, from 36 to 54 months of follow-up, no additional persons in the former group experienced ESKD.¹³⁹ Of interest is that a small number of persons who continued taking ACE inhibitors exhibited an increase in GFR after prolonged treatment.¹⁴⁰

One RCT confirmed that the renoprotective benefits of ACE inhibitor treatment may be observed even in advanced stages of CKD. Among 244 persons with a serum creatinine level of 3.1 to 5.0 mg/dL at baseline, random assignment to ACE inhibitor treatment was associated with a 52% reduction in urine protein levels and a 43% reduction in the risk of the primary endpoint (serum creatinine doubling, ESKD, or death).¹⁴¹ A meta-analysis of 11 studies that included 1860 persons with nondiabetic CKD¹⁴² revealed that ACE inhibitor treatment was associated with significantly lower risks of ESKD (RR, 0.69; 95% CI, 0.51–0.94) and with the composite endpoint of serum creatinine doubling or ESKD (RR, 0.70; 95% CI, 0.55–0.88). Moreover, the benefits of ACE inhibitor treatment were greater in persons with more severe baseline proteinuria but were inconclusive in persons with urine protein levels of less than 0.5 g/day. A further analysis restricted to persons with autosomal dominant polycystic kidney disease showed greater reduction in proteinuria with ACE inhibitor treatment, but overall evidence of slowing CKD progression was inconclusive and was limited to persons with more severe proteinuria.¹⁴³

In addition to the renoprotective benefits of ACE inhibitor treatment, the HOPE study reported substantial reductions in overall mortality (RR, 0.84) and cardiovascular mortality (RR, 0.74) among 9297 participants who were at increased risk of cardiovascular disease receiving an ACE inhibitor versus placebo.^{142a} Although the HOPE study did not include large numbers of persons with nondiabetic CKD, cardiovascular disease remains the most widespread cause of morbidity and mortality among these persons, and the data therefore provide further support for the use of ACE inhibitor therapy in persons with CKD.

In view of the unequivocal data regarding renoprotection and the probable reduction in cardiovascular risk, we recommend ACE inhibitor treatment for all persons with CKD and urine protein levels greater than 0.5 g/day (ACR >30 mg/mmol; PCR >50 mg/mmol) unless there are specific contraindications. The KDIGO guidelines similarly recommend treatment with an ACE inhibitor or ARB in adults with CKD and urine albumin excretion exceeding 300 mg/day (or equivalent).⁵

ANGIOTENSIN RECEPTOR BLOCKERS

Diabetic Kidney Disease

ARBs inhibit the RAAS by blocking angiotensin II subtype 1 receptors. Although ACE inhibitors and ARBs differ significantly in their effects on the RAAS in ways that may be therapeutically relevant, experimental studies indicate that

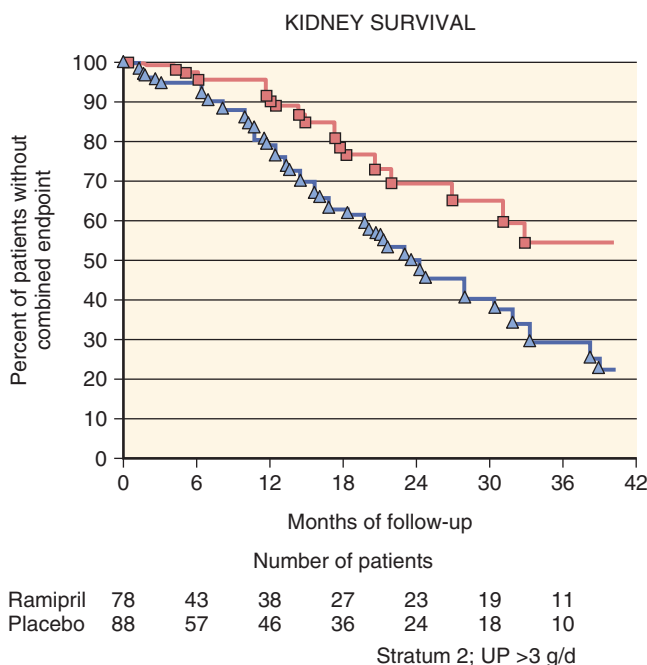


Fig. 59.7 Kaplan–Meier plot showing the cumulative incidence of the primary endpoint of doubling of serum creatinine level or end-stage kidney disease in persons with nondiabetic chronic kidney disease and urine protein levels exceeding 3 g/day. Persons were randomly assigned to receive treatment with ramipril (squares) or placebo (triangles); the graph shows improved renal survival in the ramipril recipients. (From Gruppo Italiano di Studi Epidemiologici in Nefrologia [GISEN]. Randomised placebo-controlled trial of effect of ramipril on decline in glomerular filtration rate and risk of terminal renal failure in proteinuric, non-diabetic nephropathy. *Lancet* 1997;349:1857–1863.)

both treatments produce similar changes in glomerular hemodynamics (for a given blood pressure change) and afford equivalent renoprotection in a variety of CKD models.^{143a} Three large RCTs published simultaneously established a clear role for ARB therapy in achieving renoprotection for persons with type 2 diabetes. In the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) trial, 1513 persons with overt diabetic nephropathy were randomly assigned to receive ARB treatment or placebo and were monitored for a mean of 3.4 years.¹⁴⁴ ARB treatment was associated with significant reductions in the incidence of serum creatinine doubling (RR reduction, 25%) and in ESKD (RR reduction, 28%; Fig. 59.8).

In IDNT, 1715 persons with overt diabetic nephropathy were randomly assigned to receive treatment with the ARB irbesartan, amlodipine, or placebo.¹⁴⁵ After a mean of 2.6 years, the risk of serum creatinine doubling was 33% lower with irbesartan than with placebo and 37% lower with irbesartan than with amlodipine. ARB treatment was associated with a 23% reduction in the risk of ESKD in comparison with placebo and amlodipine, but this reduction was not statistically significant (Fig. 59.9). Of importance is that close

matching of achieved blood pressure between groups in both these trials implies that, as with the ACE inhibitor studies, the additional renoprotective effects of ARB treatment could not be attributed merely to their antihypertensive effects.

In a third study, investigators examined the renoprotective effects of an ARB (irbesartan) in 590 persons with type 2 diabetes, hypertension, and microalbuminuria.¹⁴⁶ Persons were randomly assigned to receive irbesartan at one of two different dosages (300 or 150 mg/day) or placebo. After 2 years, there were significant differences in the incidence of overt proteinuria (5.2%, 9.7%, and 14.9%), and the higher dose of irbesartan was associated with substantial reduction in the risk of overt nephropathy (HR, 0.30; 95% CI, 0.14–0.61) in comparison with placebo. This dose-dependent effect indicates that when ARBs are used to treat diabetic microalbuminuria, the dose should be titrated up to the maximum antihypertensive dose.

A meta-analysis confirmed the results of individual trials by showing significant reductions in the risk of ESKD (RR, 0.78; 95% CI, 0.67–0.91) and in doubling of serum creatinine level (RR, 0.79; 95% CI, 0.67–0.93), as well as a reduction in risk of progression from microalbuminuria to macroalbuminuria

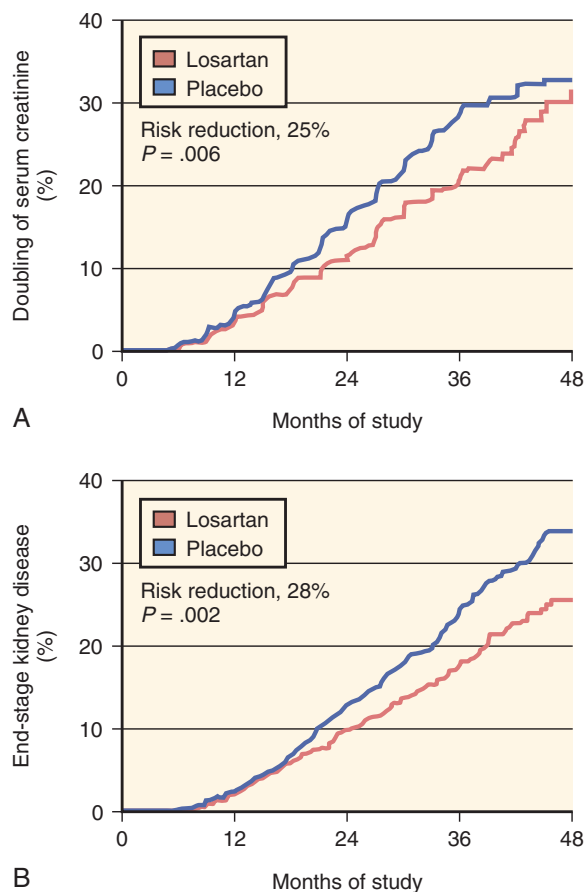


Fig. 59.8 Kaplan-Meier curves showing the incidence of (A) doubling of serum creatinine level and (B) end-stage kidney disease in persons with type 2 diabetes and nephropathy who were randomly assigned to receive treatment with losartan or placebo. (From Brenner BM, Cooper ME, de Zeeuw D, et al. Effects of losartan on renal and cardiovascular outcomes in persons with type 2 diabetes and nephropathy. *N Engl J Med.* 2001;345:861–869.)

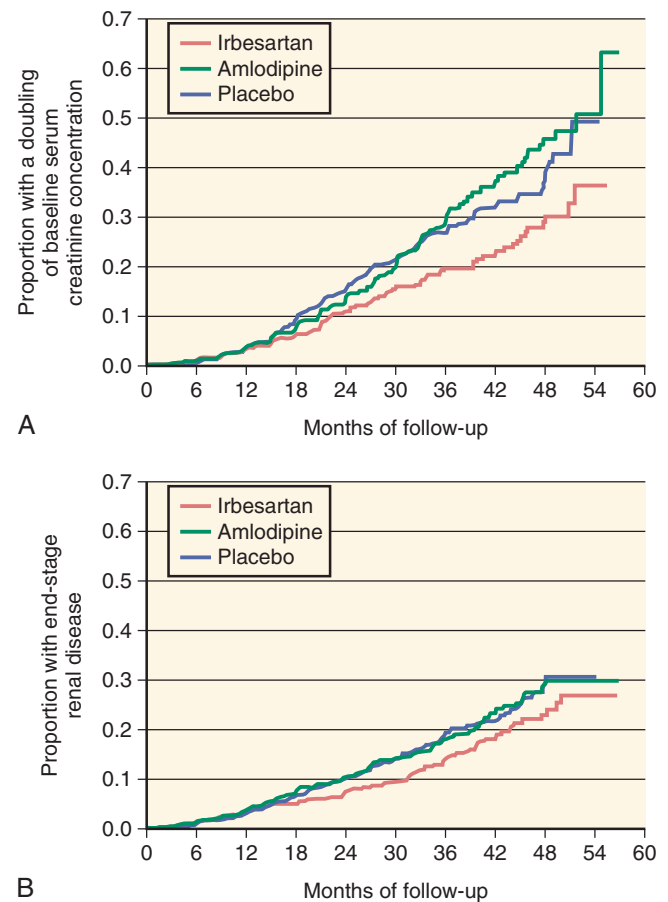


Fig. 59.9 Kaplan-Meier curves showing the incidence of (A) doubling of serum creatinine levels and (B) end-stage kidney disease in persons with type 2 diabetes and nephropathy who were randomly assigned to receive treatment with irbesartan, amlodipine, or placebo. (From Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in persons with nephropathy due to type 2 diabetes. *N Engl J Med.* 2001;345:851–860.)

(RR, 0.49; 95% CI, 0.32–0.75) among diabetic persons treated with ARB versus placebo.¹³⁴ Of interest is that there was no reduction in all-cause mortality.

Several RCTs have been conducted to investigate the potential role of ARB treatment to prevent the development of microalbuminuria in persons with diabetes. In the Renin Angiotensin System Study (RASS), normotensive persons with type 1 diabetes randomly assigned to receive treatment with losartan evidenced a significantly higher incidence of microalbuminuria than did the participants receiving placebo (17% vs. 6%; $P = .01$),¹³² and in the Diabetic Retinopathy Candesartan Trials (DIRECT), mainly normotensive persons with type 1 or type 2 diabetes randomly assigned to receive treatment with candesartan evidenced no reduction in the incidence of microalbuminuria in comparison with participants receiving placebo.¹⁴⁷ By contrast, when hypertensive persons with type 2 diabetes were studied in the Randomized Olmesartan and Diabetes Microalbuminuria Prevention (ROADMAP) trial, those who received treatment with olmesartan evidenced a 23% delay in the time to onset of microalbuminuria in comparison with participants who received placebo. A small but significantly higher incidence of death from cardiovascular causes was observed, particularly in persons with a previous history of cardiovascular disease and in those with the greatest reduction in blood pressure.¹⁴⁸

In summary, there is sufficient evidence to support the use of ARB treatment to achieve renoprotection in persons with type 2 diabetes and overt nephropathy. ARB treatment is also effective in preventing progression from microalbuminuria to overt diabetic nephropathy in persons with type 2 diabetes. Data from the ROADMAP trial show that ARB treatment delays the onset of microalbuminuria in hypertensive persons with type 2 diabetes, but the associated increase in death from cardiovascular disease remains a concern. It seems reasonable to recommend ARB treatment for hypertensive persons with type 2 diabetes in the absence of cardiovascular disease. By contrast, no renoprotective benefit is associated with ARB treatment in normotensive persons with type 1 or 2 diabetes. The KDIGO guidelines recommend treatment with an ACE inhibitor or ARB in all adults with diabetes and urine albumin excretion 30 mg/day or higher (or equivalent).⁵

Nondiabetic Chronic Kidney Disease

No large RCTs have investigated the renoprotective effects of ARBs versus other antihypertensive drugs in nondiabetic kidney disease. As will be discussed later, published evidence indicates that it is reasonable to expect that ARBs will provide similar renoprotective benefits in nondiabetic CKD to those observed in diabetic kidney disease. Based on the evidence, we recommend ACE inhibitor treatment as first-line therapy for nondiabetic, proteinuric CKD but ARBs provide a valuable alternative in persons who are intolerant to ACE inhibitors. The KDIGO guidelines recommend treatment with an ACE inhibitor or ARB in adults with all forms of CKD and albuminuria greater than 300 mg/day (or equivalent).⁵

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS VERSUS ANGIOTENSIN RECEPTOR BLOCKER TREATMENT

Despite differences in their mode of action, the renoprotective effects of ACE inhibitors and ARB treatment have been directly

compared in few studies. In a mixed group of persons with type 2 diabetes and microalbuminuria or macroalbuminuria, there was no significant difference between ACE inhibitors and ARB treatment in the primary outcome of change in GFR or a secondary outcome of UACR.¹⁴⁹ According to a meta-analysis of small trials in which ACE inhibitors were compared with ARB treatment in persons with diabetic CKD, benefits were similar with regard to incidence of ESKD, serum creatinine doubling, and progression of microalbuminuria to macroalbuminuria.¹⁵⁰ Similarly, in ONTARGET, there was no difference in cardiovascular or renal endpoints between persons randomly assigned to receive treatment with an ACE inhibitor and those assigned to receive an ARB.¹⁵¹ By contrast, in the RASS, ARB treatment was associated with a higher incidence of microalbuminuria than was placebo in persons with type 1 diabetes, whereas ACE inhibitors were not significantly different from placebo in this regard.¹³²

Most national and international guidelines therefore recommend ACE inhibitor or ARB treatment for all forms of diabetic and nondiabetic CKD and leave the choice to individual physicians. One advantage of ARBs over ACE inhibitors is their more favorable side-effect profile.¹⁵² In clinical trials, ARBs have been reported to have side-effect profiles similar to those of placebo^{153,154} and, in particular, are not associated with the cough that may occur in up to 20% of persons receiving an ACE inhibitor. Among persons who switched from ACE inhibitors to ARB therapy, cough recurred significantly less often than in persons rechallenged with an ACE inhibitor.^{155,156}

In choosing between ACE inhibitor and ARB therapy for persons with type 2 diabetes and diabetic nephropathy, physicians have to consider evidence of proven renoprotection with ARB treatment versus a mortality benefit associated with ACE inhibitor treatment (in persons without established diabetic nephropathy). One meta-analysis suggested a modestly increased risk of new cancer occurrence in persons on ARB treatment; this finding has created some controversy.¹⁵⁷ However, two long-term antihypertensive cohort studies linked to cancer registries, which can explore competing risk, showed no increased cancer risk from long-term hypertension therapy and RAAS blockade.^{158,159} A large network meta-analysis of 70 RCTs of antihypertensive therapy in 324,168 persons recorded no difference in risk of cancer between the participants receiving treatment (including ARBs) and control participants.¹⁶⁰

Some meta-analyses have called into question the value of RAAS inhibition for renoprotection. In one study, trials of ACE inhibitor and ARB treatment were pooled, and when studies of diabetic and nondiabetic CKD were considered together, the analysis revealed a benefit in reducing the risk of ESKD (RR, 0.87; 95% CI, 0.75–0.99) and albuminuria (mean urine albumin level, –15.7 mg/day; 95% CI, –24.7 to –6.7 mg/day) but no significant benefit in reducing the risk of serum creatinine doubling (RR, 0.71; 95% CI, 0.49–1.04). When data from studies of diabetic and nondiabetic CKD were considered separately, no significant benefits were evident with regard to incidence of ESKD or serum creatinine doubling, but the benefit for albuminuria reduction persisted. The authors concluded that the renoprotective effects of ACE inhibitor or ARB therapy probably result only from their antihypertensive effects.¹⁶¹ This meta-analysis is, however, flawed in our opinion, and its conclusions have been rejected

by many other investigators.¹⁶² The principal weaknesses were inclusion of data from the large ALLHAT study, in which only a minority of persons (5662 of 33,357) actually had CKD⁸⁸; heterogeneity across trials that should not have been pooled; and lack of patient-level data.¹⁶³ A more recent meta-analysis focused on the potential benefits of RAAS inhibitors in persons with diabetes. The analysis included 19 RCTs that enrolled 25,414 participants and reported no significant difference between groups receiving RAASi inhibitors or other antihypertensives for risk of death, cardiovascular death, and CVE apart from heart failure, which was reduced with RAAS inhibitor treatment (HR, 0.78; 95% CI, 0.70–0.88). In addition, no difference was observed in incidence of ESKD, the only renal outcome evaluated.¹⁶⁴ These findings should, however, be interpreted in light of several substantial limitations. First, only a minority of the included trials enrolled participants with CKD and the findings are therefore not necessarily applicable to persons with diabetes and CKD. Second, median follow-up was relatively short at 3.8 years, which is too short to expect any impact of RAAS inhibitor treatment on the incidence of ESKD in the populations studied. Third, the analysis excluded placebo-controlled trials, ignoring the fact that in many of these trials other antihypertensive medication was prescribed to achieve similar control of blood pressure between groups.

By contrast, a large-network meta-analysis included data from 43,256 participants with diabetes and CKD from 157 randomized trials that compared one blood pressure treatment with another or placebo or control. No drug treatment was more effective than placebo in reducing all-cause mortality but ARB treatment was associated with a significant reduction in incidence of ESKD (OR, 0.77; 95% CI, 0.65–0.92) and combination ACE inhibitor and ARB was similarly effective (OR, 0.62; 95% CI, 0.43–0.90). The effect of ACE inhibitor monotherapy was of borderline statistical significance (OR, 0.71; 95% CI, 0.51–1.01). Endothelin inhibitors, ACE inhibitors, and ARBs significantly reduced the risk of serum creatinine doubling. All regimens that included an ACE inhibitor or ARB were associated with regression of albuminuria. No treatment significantly increased the risks of hyperkalemia or AKI, though for combination ACE inhibitor and ARB therapy this was of borderline significance. Interestingly some other treatments were associated with adverse outcomes when used as monotherapy: Calcium channel blockers did not improve albuminuria and tended to increase doubling of serum creatinine; β -blockers were associated with reduced survival; renin inhibitors increased the risk of serum creatinine doubling.¹⁶⁵

COMBINATION TREATMENT WITH ANGIOTENSIN-CONVERTING ENZYME INHIBITORS AND ANGIOTENSIN RECEPTOR BLOCKERS

The differing mechanisms of ACE inhibitors and ARBs on the RAAS imply that in combination, they may have additive or synergistic effects. The added antihypertensive effects of combination therapy have, however, made it difficult to separate the benefits of additional blood pressure lowering from added renoprotection directly attributable to dual blockade of the RAAS. In several small studies, researchers reported additional lowering of urine protein excretion with combination therapy of ACE inhibitors and ARBs, in comparison with monotherapy; when pooled in three meta-analyses, all

data indicate that there is greater antiproteinuric effect with combination RAAS inhibition therapy.^{166–168} Only one RCT yielded results indicating benefit associated with ACE inhibitor and ARB combination therapy in CKD with respect to serum creatinine doubling or ESKD incidence. Publication of that study was, however, withdrawn because of concerns about the conduct of the study and integrity of the data.¹⁶⁹

Furthermore, the results of the ONTARGET trial cast doubt on the benefit of dual therapy.¹⁵¹ In ONTARGET, 25,620 persons with hypertension and additional cardiovascular risk factors were randomly assigned to receive therapy with an ACE inhibitor alone, an ARB alone, or a combination of both.¹⁷⁰ The primary analysis revealed no difference in CVEs between the randomized groups, but the number of events for the composite renal outcome—need for dialysis, doubling of serum creatinine level, and death¹⁵¹—was increased with combination therapy ($n = 1233$, or 14.5%; HR, 1.09; 95% CI, 1.01–1.18; $P = .037$) versus monotherapy (ARB: $n = 1147$, or 13.4%; ACE inhibitor: $n = 1150$, or 13.5%). This excess was attributable predominantly to more acute dialysis and to the combination of all types of dialysis and serum creatinine doubling. In addition, hyperkalemia ($K > 5.5$ mmol/L) was more frequent with combination therapy (ACE inhibitor, 3.2%; ARB, 3.3%; combination, 5.6%; $P < .001$ for combination versus ACE inhibitor). However, participants were recruited on the basis of cardiovascular risk profile, and the majority did not have reduced GFR (mean baseline eGFR, 74 mL/min/1.73 m²) or proteinuria (13% had microalbuminuria and 4% had macroalbuminuria). In addition, participants were elderly (mean age 66.5 years) and predominantly male (73%). They had substantial vascular disease: coronary artery disease (75%), previous myocardial infarction (50%), angina (35%), unstable angina (15%), and peripheral vascular disease (13%). Angiographic studies¹⁷¹ suggest that approximately 10% to 15% of this cohort would be expected to have atheromatous renal vascular disease in large vessels, and many more would have small vessel disease.

Several other studies have provided evidence of the potential risks of combination therapy. A population-based observational study reported a significantly higher incidence of the combined endpoint of serum creatinine doubling, ESKD, or death in elderly persons (older than 65 years) receiving combination therapy with ACE inhibitors and ARBs than among those receiving monotherapy (5.2 vs. 2.4 events per 1000 persons per month; adjusted HR, 2.36; 95% CI, 1.51–3.71). The incidence of hyperkalemia was also higher with combination therapy (2.5 vs. 0.9 events per 1000 persons per month; adjusted HR, 2.42; 95% CI, 1.36–4.32).¹⁷² A further RCT (VA NEPHRON-D) in 1448 people with type 2 diabetes and UACR of at least 300 mg/g was stopped early due to safety concerns. Analysis of available data found no benefit with respect to the primary outcome of CKD progression, ESKD, or death in participants randomized to combination versus monotherapy, though early termination of the trial reduced the potential to observe benefit. Of concern, those receiving combination therapy evidenced a significantly higher incidence of AKI (12.2 vs. 6.7 events per 100 person-years; $P < .001$) and hyperkalemia (6.3 vs. 2.6 events per 100 person-years; $P < .001$).¹⁷³

In a meta-analysis of 59 RCTs comparing the efficacy and safety of combination versus single RAAS inhibitor therapy in CKD, combination therapy was associated with significant

improvement in urine albumin and protein excretion as well as blood pressure control. These beneficial effects, however, were associated with a net 1.8 mL/min/1.73 m² decline in GFR, a significant increase in serum potassium level (3.4% higher rate of hyperkalemia), and a 4.6% higher rate of hypotension. There was no effect on doubling of the serum creatinine level, hospitalization, or mortality.¹⁷⁴

The KDIGO guidelines do not recommend the use of combination ACE inhibitor and ARB therapy for renoprotection⁵ and NICE guidelines⁷⁸ as well as the 2017 American College of Cardiology/American Heart Association guidelines¹¹⁵ recommend against combination therapy. Nevertheless, as discussed earlier, there is some evidence of potential benefit, particularly in nondiabetic, proteinuric CKD and it may be possible to reduce the risk of harm by avoiding combination therapy in high-risk groups, such as those with extensive atherosclerosis. It has been proposed that using lower doses of ACE inhibitor and ARB in combination may achieve benefit while reducing harm and this hypothesis is currently being tested in an RCT.

ALDOSTERONE ANTAGONISM

Aldosterone has been identified as an important factor in the pathogenesis of hypertension, cardiovascular disease, and progressive kidney injury through hemodynamic and profibrotic actions. Treatment with spironolactone and other aldosterone antagonists has produced renoprotective effects in experimental¹⁷⁵ and small clinical studies.¹⁷⁶ One meta-analysis of their use in comparison with other RAAS-inhibiting agents included data from 10 trials and 845 participants. In comparison with ACE inhibitor or ARB plus placebo, nonselective aldosterone antagonists added to ACE inhibitor or ARB treatment significantly reduced proteinuria (weighted MD, -0.80 g/day; 95% CI, -1.23 to -0.38) and blood pressure, but these developments did not translate into an improvement in GFR (weighted MD, -0.70 mL/min/1.73 m²; 95% CI, -4.73 to 3.34). There was a significant increase in the risk of hyperkalemia with the addition of an aldosterone antagonist (RR, 3.06; 95% CI, 1.26–7.41).¹⁷⁷ A more recent meta-analysis of 27 studies confirmed that treatment with a nonselective aldosterone antagonist (spironolactone) alone or in combination with an ACE inhibitor or ARB was associated with reduction in proteinuria (standardized MD, -0.61 g/day; 95% CI, -1.08 to -0.13) and blood pressure (MD, -3.44 mm Hg; 95% CI, -5.05 to -1.83) but provided no data on reduction in CVEs or ESKD. In addition, aldosterone antagonist treatment was associated with an increased risk of hyperkalemia (RR, 2.00; 95% CI, 1.25–3.20) and gynecomastia compared with ACE inhibitor or ARB (or both) (RR, 5.14; 95% CI, 1.14–23.23).¹⁷⁸ The renoprotective potential of the more novel nonsteroidal aldosterone antagonists is also starting to be evaluated. In one RCT that enrolled 336 persons with hypertension, UACR 30 to 599 mg/g, and eGFR of 50 mL/min/1.73 m² or above, treatment with eplerenone for 52 weeks, added to ACE inhibitor of ARB therapy, resulted in a 17.3% decrease in UACR, whereas a 10.3% increase was observed in those who received placebo ($P = .02$). Serum potassium was higher with eplerenone treatment but no episodes of severe hyperkalemia (serum potassium > 5.5 mmol/L) were observed.¹⁷⁹

Aldosterone antagonists are recommended as part of the management of heart failure and resistant hypertension, but

their use is often limited by the development of hyperkalemia, particularly in persons with reduced GFR. The use of novel potassium-binding agents has been proposed as a strategy to prevent hyperkalemia and facilitate aldosterone antagonist treatment and clinical trials are underway to evaluate this in the context of CKD.¹⁸⁰ Until the results of further RCTs are published, treatment with an aldosterone antagonist cannot be recommended for the management of CKD, though it may be beneficial to improve control of resistant hypertension, provided that care is taken to monitor and avoid hyperkalemia. Further studies are required to evaluate the long-term effects of aldosterone antagonists on renal outcomes, mortality, and safety.

DIRECT RENIN INHIBITORS

Direct renin inhibitors inhibit the RAAS at its rate-limiting step (the conversion of angiotensinogen to angiotensin I) and may therefore achieve more complete blockade of the RAAS than do ACE inhibitors or ARBs. Direct renin inhibitors are effective antihypertensive drugs and reduced proteinuria in experimental models of CKD^{181,182} and RCTs.^{183,184} However, a large randomized trial that included 8561 people with type 2 diabetes and albuminuria or cardiovascular disease was stopped prematurely after the second interim efficacy analysis. Despite greater lowering of blood pressure and albuminuria with a combination of direct renin inhibitor and ARB therapy, no benefit was observed with respect to the composite primary endpoint of CVE, CKD progression, ESKD, or death versus ARB monotherapy, but combination therapy was associated with a higher incidence of hyperkalemia and hypotension.^{185,186} There is currently, therefore, no evidence that direct renin inhibitor therapy, alone or in combination with ACE inhibitor therapy, affords more effective renoprotection than ACE inhibitor or ARB therapy alone.

Clinical Relevance

Antihypertensive therapy with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers as “first line” in persons with albuminuria is the mainstay of treatment to achieve optimal renoprotection. Treatment should be started at a low dose and titrated upward. Serum creatinine and potassium should be checked approximately 1 week after initiation or dose titration.

SODIUM GLUCOSE COTRANSPORTER 2 (SGLT2) INHIBITORS

The sodium glucose cotransporter 2 (SGLT2) inhibitors block the reabsorption of glucose and sodium in the proximal tubule and were developed primarily as a treatment for type 2 diabetes mellitus. RCTs in persons with diabetes and normal or moderately reduced GFR found that these drugs improved glycemic control but importantly also afforded significant reductions in fatal and nonfatal CVE, all-cause mortality, and multiple measures of diabetic kidney disease progression.¹⁸⁷ These renoprotective effects have been attributed to multiple mechanisms including lowering of system blood pressure and intraglomerular hydraulic pressure, amelioration of

glomerular hyperfiltration, reduction of albuminuria, weight loss, amelioration of arterial stiffness, and induction of hypoxia-inducible factor 1.¹⁸⁸ At present, the SGLT2 inhibitors are indicated only in persons with type 2 diabetes and normal or mildly reduced GFR (eGFR ≥ 45 mL/min/1.73 m² for some drugs) but further trials are investigating whether these drugs may afford renal protection in persons with more advanced CKD. The CREDENCE trial enrolled 4401 adults with type 2 diabetes, eGFR 30 to 89 mL/min/1.73 m², and UACR 300 to 5000 mg/g despite treatment with the maximum dose of an ACE inhibitor or ARB. The study was stopped early due to evidence of benefit in a planned interim analysis. The risk of the composite primary outcome of ESKD, doubling of serum creatinine, and cardiovascular or renal death, was reduced by 30% in participants who received canagliflozin (HR, 0.70; 95% CI, 0.53 to 0.81; $p < 0.001$) and the risk of ESKD was reduced by 32% (HR, 0.68; 95% CI, 0.54 to 0.86; $p = 0.002$).¹⁸⁹ Furthermore, trials are now underway to evaluate the renoprotective potential of SGLT2 inhibitors across a broader spectrum of CKD including nondiabetic CKD. The SGLT2 inhibitors are reviewed in more detail in relation to the treatment of diabetic kidney disease in Chapter 39.

HYPERURICEMIA

A number of studies have highlighted evidence that elevated serum uric acid concentration is a risk factor for the development and progression of CKD.^{190,191} It has also been associated with excess cardiovascular risk and hypertension in a cohort of community-based persons.¹⁹² Whether it is an independent risk factor for these outcomes in CKD or a marker of more severe renal and cardiovascular outcomes is an issue of ongoing debate.¹⁹³ In animal studies, elevated serum uric acid decreases nitric oxide production, which provokes endothelial dysfunction, increases blood pressure, promotes fibrosis, and causes the release of proinflammatory cytokines and thereby results in T-cell activation. A meta-analysis of eight small trials investigating the potential benefits of lowering uric acid in CKD with allopurinol included data from 476 participants. In three studies that reported serum creatinine concentration, allopurinol treatment was associated with a lower serum creatinine than was placebo, but in five studies that reported eGFR, there was no significant difference between groups treated with allopurinol or placebo. In five trials that measured proteinuria, no benefit was observed. Progression to ESKD was reported in only two studies and was not different between treatment groups, although the incidence was too low to allow robust conclusions to be made.¹⁹⁴ A more recent meta-analysis included nine trials of all xanthine oxidase inhibitors (febuxostat and topiroxostat in addition to allopurinol) in persons with CKD and hyperuricemia.¹⁹⁵ In three studies the risk of the combined outcome of 50% reduction in eGFR, doubling of serum creatinine, or initiation of dialysis was significantly reduced by xanthine oxidase inhibitor treatment (204 participants; RR, 0.42; 95% CI, 0.22–0.80). Better preservation of eGFR was observed in data pooled from RCTs of minimum 3 months' duration (4 studies, 357 patients; MD, 6.82 mL/min/1.73 m²; 95% CI, 3.50–10.15) and in trials of high methodological quality (blind design; 3 studies, 400 patients; MD, 2.61 mL/min/1.73 m²; 95% CI, 0.23–4.99). However, no benefits were observed with respect to serum creatinine or proteinuria.¹⁹⁵

One randomized trial of allopurinol in 113 persons with CKD (included in the aforementioned meta-analysis) reported a 47% reduction in risk of CKD progression (defined as decrease in eGFR of >0.2 mL/min/month) in participants treated with allopurinol after multivariable Cox proportional hazards analysis, as well as a 71% reduction in risk of new CVEs and 62% reduction in risk of hospitalization.¹⁹⁶ Long-term follow-up of these participants reported further benefit with a 68% reduction in risk of a renal event (initiation of RRT or 50% reduction in GFR or doubling of serum creatinine) and a 57% reduction in risk of CVE.¹⁹⁷ One observational study compared outcomes in 874 persons with CKD and hyperuricemia treated with benzbromarone, allopurinol, or febuxostat using Cox proportional hazards models. Febuxostat was more effective in lowering serum uric acid concentration than the other treatments. Benzbromarone was associated with a lower risk of ESKD (defined by initiation of dialysis) than allopurinol (HR, 0.50; 95% CI, 0.25–0.99), which evidenced similar lack of benefit to febuxostat. Participants who achieved the target for lowering serum uric acid concentration with febuxostat or benzbromarone also had a reduced risk of ESKD.¹⁹⁸

Despite the aforementioned evidence suggesting renoprotective benefit, the trials published to date are of insufficient size to recommend uric acid lowering as part of the management of CKD unless there is clinical evidence of gout.

TREATMENT OF METABOLIC ACIDOSIS

As the number of functioning nephrons declines, CKD leads to net retention of hydrogen ions, which begins when GFR falls below 40 to 50 mL/min/1.73 m².¹⁹⁹ Among persons in whom GFR decreases from 90 to less than 20 mL/min/1.73 m², the prevalence of metabolic acidosis rises from 2% to 39% and is higher among younger persons and those with diabetes.²⁰⁰ As the patient approaches ESKD, the plasma bicarbonate concentration tends to stabilize between 15 and 20 mEq/L. Chronic metabolic acidosis has multiple adverse consequences, including increased protein catabolism, increased bone turnover, induction of inflammatory mediators, insulin resistance, and increased production of corticosteroids and parathyroid hormone. Several observational studies, including the CRIC study, have identified low serum bicarbonate as a risk factor for CKD progression,²⁰¹ although post hoc analysis of data from persons with diabetic kidney disease enrolled in the RENAAL and IDNT studies found that an association between lower serum bicarbonate and renal outcomes was not maintained after adjustment for baseline GFR.²⁰²

The first study to show convincing renoprotection with bicarbonate supplementation was from a single center and involved 134 persons with advanced CKD (creatinine clearance rates between 15 and 30 mL/min/1.73 m²) and baseline serum bicarbonate concentrations of 16 to 20 mEq/L. The participants were randomly assigned to receive treatment with oral bicarbonate or no treatment.²⁰³ After 2 years of follow-up, there was a lower mean rate of decline in creatinine clearance (1.88 vs. 5.93 mL/min/1.73 m²) and a lower risk of ESKD among the persons who received the bicarbonate treatment than among the controls (6.5% vs. 33%). In a subsequent randomized, placebo-controlled trial in persons with a mean eGFR of 75 mL/min/1.73 m², treatment with sodium bicarbonate for 5 years was associated with a slower rate of decline

in eGFR (derived from plasma cystatin C measurements) compared with placebo or treatment with sodium chloride.²⁰⁴ Western diets are typically acid producing, but the addition of significant portions of fruits and vegetables can move this to a base-producing state. Further studies have reported that correction of acidosis with a diet rich in fruits and vegetables was as effective as sodium bicarbonate in ameliorating kidney damage in early (stage 1 or 2 CKD)²⁰⁵ and more advanced (stage 4 CKD) disease.²⁰⁶ Furthermore, a recent trial has reported benefit even in people with mild acidosis. People with stage 3 CKD and serum bicarbonate 22 to 24 mmol/L were randomized to oral bicarbonate supplementation, a diet rich in fruits and vegetables, or “usual care.” All participants received treatment with an RAAS inhibitor, and SBP was controlled to lower than 130 mm Hg. Both interventions achieved an increase in serum bicarbonate and were associated with a decrease in urinary angiotensinogen. After 3 years, both interventions were associated with less albuminuria and GFR decline than the “usual care” group.²⁰⁷ The KDIGO guidelines recommend bicarbonate supplementation for persons with levels below 22 mEq/L,³ but further studies are required to further investigate whether this may also be beneficial in the setting of less severe acidosis.

PROTEINURIA AS A THERAPEUTIC TARGET

Proteinuria is a marker of glomerular filtration barrier integrity, and the magnitude of proteinuria has therefore been used as an indicator of the severity of glomerulopathy. This view has been confirmed by several observations that the severity of proteinuria at baseline is the most important independent predictor of renal outcomes in randomized trials of persons with diabetic nephropathy^{208,209} and nondiabetic nephropathy.^{210–212} In addition, it has been proposed that proteinuria per se contributes to progressive renal injury²¹³ and that amelioration of proteinuria should therefore be viewed as a therapeutic goal.

Support for this hypothesis is derived from a number of sources. In the MDRD study, reduction in urine protein levels, independent of blood pressure, was associated with slower progression of CKD, and the degree of benefit achieved through blood pressure lowering was dependent on the extent of baseline proteinuria.⁹⁹ Furthermore, several other investigators have observed that the percentage reduction in urine protein level after initiation of ACE inhibitor or ARB treatment and the magnitude of proteinuria during treatment (residual proteinuria) are strong independent predictors of the subsequent rate of decline in GFR among persons with diabetic and nondiabetic CKD.^{208,209} A meta-analysis that included data from 1860 persons with nondiabetic CKD confirmed these findings and showed that during antihypertensive treatment, the achieved level of urine protein was a powerful predictor of the combined endpoint of serum creatinine doubling or onset of ESKD (RR, 5.6 for each 1.0-g/day increase in achieved level of proteinuria).¹⁴² A further meta-analysis of 21 randomized trials of drug treatment in CKD that included 78,342 participants found that for each 30% initial reduction in albuminuria on treatment, the risk of ESKD decreased by 23.7% (95% CI, 11.4%–34.2%) independent of the class of drug used for treatment.²¹⁴

One RCT provided direct evidence that the extent to which proteinuria is lowered determines subsequent prognosis. In

the Renoprotection of Optimal Antiproteinuric Doses (ROAD) study, 360 nondiabetic persons with proteinuria and CKD were randomly assigned to receive a conventional dosage of benazepril (10 mg/day), a conventional dosage of losartan (50 mg/day), an upward titration of benazepril (range of 10–40 mg/day), or an upward titration of losartan (range of 50–200 mg/day).²¹⁵ In upward titration, the dosage of the RAAS-inhibiting agent was increased to maximize the anti-proteinuric effect. After a median of 3.7 years, titration of benazepril and losartan to the maximum antiproteinuric dose, in comparison with the fixed conventional dosages, reduced the risk of serum creatinine doubling by 49% and 50%, respectively, and the risk of ESKD by 47% in both groups (Fig. 59.10). Both agents provided similar overall RR reductions at optimal antiproteinuric dosages. Reduction of urine protein levels at 3 months (approximately 50% decrease with the upward-titration strategy) was closely correlated with the subsequent rate of GFR decline. Participants in all groups of the study had similar reductions in blood pressure. Regardless of whether proteinuria contributes directly to renal injury, the strong association between achieved reduction in urine protein and renoprotection in clinical studies implies that amelioration of proteinuria should be regarded as an important therapeutic goal in renoprotective strategies. We recommend a goal of reducing proteinuria to less than 0.5 g/day (equivalent UACR of $\approx$ 300 mg/g or 30 mg/mmol).

Clinical Relevance

Proteinuria should be regarded as a therapeutic target and renin–angiotensin–aldosterone system inhibitor therapy should be escalated and other antihypertensive therapy added to achieve blood pressure targets with the aim of reducing proteinuria to less than 0.5 g/day.

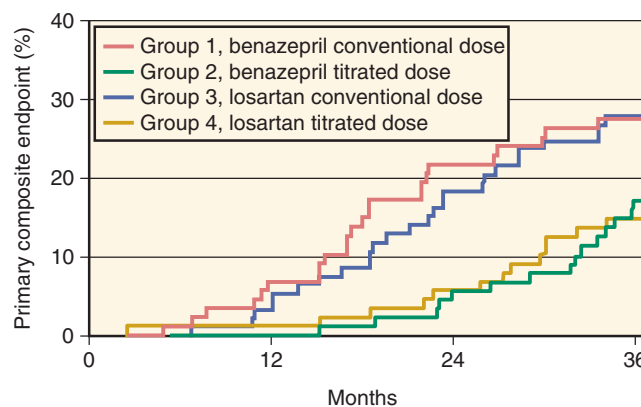


Fig. 59.10 Kaplan-Meier curves showing the incidence of the combined endpoint of doubling of serum creatinine level or end-stage kidney disease in persons with nondiabetic chronic kidney disease and urine protein levels exceeding 1 g/day. Persons in group 1 were randomly assigned to receive standard doses of angiotensin-converting enzyme (ACE) inhibitors; those in group 3 received standard doses of angiotensin receptor blockers (ARBs); those in group 2 received ACE inhibitors titrated upward to the maximum antiproteinuric levels; and those in group 4 received ARBs titrated upward similarly. (From Hou FF, Xie D, Zhang X, et al. Renoprotection of Optimal Antiproteinuric Doses [ROAD] study: a randomized controlled study of benazepril and losartan in chronic renal insufficiency. *J Am Soc Nephrol.* 2007;18: 1889–1898.)

TIME COURSE FOR PROTEINURIA RESPONSE

In most persons, the reduction of proteinuria takes several weeks to achieve its maximal effect, which should be considered when dosages are titrated.²¹⁶ In persons in earlier stages of CKD (with diabetes and microalbuminuria, for instance) the effects are observed more promptly and tend to parallel the time course and fall in blood pressure.²¹⁷

MONITORING AND SAFETY CONSIDERATIONS

Regular monitoring is essential for optimizing therapeutic interventions to slow CKD progression and ensure safety. Kidney function is best assessed with the use of eGFR, derived from a serum creatinine measurement with the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation, recommended by KDIGO.⁵ This strategy allows direct monitoring of the rate of GFR decline and assessment of the therapeutic goal to reduce this decline to less than 1 mL/min/year, a rate associated with normal aging. The majority of laboratories now facilitate this monitoring by reporting eGFR with every serum creatinine measurement. In addition, monitoring allows for the detection of side effects of drug treatment and, in particular, of electrolyte disorders (hyperkalemia and hyponatremia), as well as acute changes in kidney function related to volume depletion.

RENAL DYSFUNCTION AND HYPERKALEMIA INDUCED BY INHIBITORS OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The appropriate frequency for monitoring of persons with CKD should depend on the CKD stage, previous rate of GFR decline, risk of future GFR decline, and use of medication that may cause acute deteriorations in GFR or electrolyte disorders (especially RAAS inhibitors and diuretics). Despite clear trial evidence of the renoprotective and cardioprotective effects of ACE inhibitors and ARBs, some physicians remain cautious about prescribing these drugs to persons with stages 3 and 4 CKD. This caution results from concerns about renal dysfunction induced by these drugs, with a potential rise in serum creatinine or potassium level (reviewed by Schoolwerth et al.²¹⁸ and Palmer²¹⁹). General guidance on risk factors for AKI and frequency of monitoring are given in [Tables 59.2](#) and [59.3](#).

The initiation of therapy with RAAS inhibition may provoke an acute decline in GFR or hyperkalemia in persons with CKD, particularly in those with volume depletion, poor cardiac status, elderly persons, those with stage 4 or 5 disease, and persons with atherosclerotic renovascular disease. GFR and electrolytes should therefore be checked before and approximately 1 week after treatment is started or dosage is increased. Based on a review (but not a meta-analysis) of data for 12 RCTs, it has previously been recommended that an acute increase in serum creatinine of up to 30% after initiation of RAAS inhibitor therapy is acceptable because it is attributable to glomerular hemodynamic effects and correlates with slower subsequent annual decline in GFR.²²⁰ This advice has recently been questioned by analysis of data from a large database of 122,363 persons initiating RAAS inhibitor therapy in primary care. The analysis confirmed that a greater than 30% increase in serum creatinine within 2 months after initiation of RAAS inhibitors (observed in 1.7% of partici-

pants) was associated with a significant increase in risks of ESKD, myocardial infarction, cardiac failure, and death, although the incidence rate ratio decreased with time after starting treatment. Importantly the study also reported a graduated relationship between all of these adverse outcomes and acute increase in serum creatinine at all values $\geq 10\%$ ([Fig. 59.11](#)).²²¹ An important limitation of this analysis is that data on proteinuria were not available. Nevertheless, these data indicate that a more cautious approach is warranted, although it is not clear whether the increase in serum creatinine is a biomarker of risk or if there is a causal relationship between the rise in serum creatinine and adverse outcomes. A rapid initial rise in serum creatinine level or a more gradual progressive increase should prompt discontinuation of therapy and consideration of further investigation to exclude renovascular disease (see [Chapter 47](#)).

It should be remembered that AKI can occur even if RAAS inhibition therapy has been successful for months or years, usually when provoked by factors such as volume depletion or nephrotoxic medications. In the Studies of Left Ventricular Dysfunction (SOLVD), a rise in serum creatinine level of 0.5 mg/dL (44 $\mu\text{mol/L}$) from baseline was noted in 16% of persons randomly assigned to receive enalapril, in comparison with 12% of persons receiving placebo. This absolute 4% excess of GFR decline was associated with older age, diuretic therapy, and diabetes. A progressive rise in serum creatinine level is much less common in persons younger than 70 years and in those without renovascular disease. As evidence for this, persons in the second Evaluation of Losartan in the Elderly (ELITE II) study of heart failure (average age of 73.5) showed relatively high rates of creatinine elevation during the 50-week follow-up period.²²² The incidence of persistent elevated serum creatinine did not differ between the losartan and captopril recipients (both 10.5%), and less than 2% of persons discontinued the RAAS agent for this reason. However, slightly more than 25% of the losartan and captopril recipients experienced at least one rise in creatinine level of 26.5 μmol (0.3 mg). These findings indicate that vigilance is needed for older persons with other comorbid conditions.

ACE inhibitor or ARB treatment should be started at a low dose and titrated upward, with monitoring of creatinine and potassium levels 5 to 7 days after each increase. To avoid compromise from intravascular volume depletion, persons should be counseled to omit ACE inhibitor or ARB treatment during vomiting or diarrheal illnesses and to seek medical advice if these illnesses do not resolve within 48 hours. Likewise, it is important to ensure adequate hydration, to omit or reduce diuretics for 48 to 72 hours if clinically appropriate, and to avoid nonsteroidal antiinflammatory drugs (NSAIDs) before starting an RAAS inhibitor. In general, we strongly advise discontinuation of NSAIDs because these are potent causes of AKI in persons with CKD.

Discontinuation of therapy because of uncontrolled hyperkalemia has been reported in only up to 4% of persons with CKD in trials, and the overall incidence was no different from that among persons taking ACE inhibitors versus non-ACE inhibitor treatment when data from six studies were combined.²²³ However, the participants were highly selected persons in trials and therefore generally have a lower risk of hyperkalemia than is observed in the general population. This important issue was highlighted by the observation

Table 59.2 Overview of Chronic Kidney Disease Management by Stage

Features	Stages 1 and 2	Stage 3a	Stage 3b	Stage 4	Stage 5
Estimated GFR	≥60 mL/min/1.73 m ² + albuminuria or hematuria or structural kidney damage	45–59 mL/ min/1.73 m ²	30–44 mL/ min/1.73 m ²	15–29 mL/min/1.73 m ²	<15 mL/min/1.73 m ²
Frequency of Monitoring	Annual	6–12 months	3–6 months	3–4 months	1–3 months
Laboratory testing	Annual electrolytes and estimated GFR Annual urine ACR (or other estimate of proteinuria) Baseline anemia and mineral and bone profiles Glucose, lipids, and HbA _{1c} See Table 59.3 for causes of AKI after initiation of ACE inhibitor or ARB therapy.				Check electrolytes and estimated GFR 1 week after new use or higher doses of ACE inhibitors or ARBs; otherwise, assess electrolytes/estimated GFR, mineral and bone, and anemia profiles every 3–6 months, depending on GFR decline.
Blood pressure targets	BP target <130/80 mm Hg with proteinuria BP target <140/90 mm Hg without proteinuria if no clinical or radiologic evidence of ARVD or previous episodes of AKI				Risk of AKI is increased in elderly persons (>75 years), those with CHF, and those with ARVD; 140/90 mm Hg may be more appropriate for these groups.
Blood pressure agents	ACE inhibitor or ARB if urine ACR ≥30 mg/g Most persons need two to four agents in total to achieve these targets, in a combination of ACE inhibitors or ARBs and one or more of the following: a diuretic (all classes), a calcium channel blocker, and a β-blocker.				Loop diuretics are now usually required for BP and edema control.
Cardiovascular prevention	Statin if CVD risk ≥10% over 10 years HbA _{1c} <7.0% unless at risk for severe hypoglycemia			Consider statin for all persons.	Continue statin.
Bone and anemia complications	If PTH level rises progressively, commence phosphate restriction and then consider therapy with vitamin D or analog. If anemia is out of keeping with GFR, confirm or rule out gastrointestinal blood loss.				Give intravenous iron before ESA if hemoglobin count <10 g/dL. Maintain target hemoglobin count of 10–11.5 g/dL.
Lifestyle and nutritional management	Smoking cessation Moderate exercise up to 30–60 min/day 4 to 7 days/wk Target weight with BMI <25 kg/m ² Reduced salt intake as per DASH diet <5 g/day				Limit dietary potassium excess. Weigh at each clinic and assess fluid overload, anorexia, physical function.
Specific RRT planning steps	Education regarding progression and role of conservative management with regard to blood pressure targets and specific primary renal disease treatment if indicated Hepatitis B vaccination if risk of progression is high			Education on RRT types and palliative care if CKD is progressing or patient is at high risk of progression. Hepatitis B vaccination.	AVF creation PD catheter insertion Enter on list for transplant
Referral guidance from primary care physician to nephrologist	Progressive or abrupt fall in estimated GFR Proteinuria (urine protein levels >0.5 g/day; ACR >300 mg/g or >30 mg/mmol)			Refer unless patient is terminally ill	Refer unless patient is terminally ill

ACE, Angiotensin-converting enzyme; ACR, albumin-to-creatinine ratio; AKI, acute kidney injury; ARB, angiotensin receptor blocker; ARVD, atherosclerotic renovascular disease; AVF, arteriovenous fistula; BMI, body mass index; BP, blood pressure; CHF, congestive heart failure; CKD, chronic kidney disease; CVD, cardiovascular disease; DASH, Dietary Approaches to Stop Hypertension; ESA, erythropoietin-stimulating agent; GFR, glomerular filtration rate; HbA_{1c}, hemoglobin A_{1c}; PD, peritoneal dialysis; PTH, parathyroid hormone; RAAS, renin-angiotensin-aldosterone system; RRT, renal replacement therapy.

Table 59.3 Causes of Acute Kidney Injury After Initiation of Therapy With Angiotensin-Converting Enzyme Inhibitor or Angiotensin Receptor Blocker**Blood Pressure Insufficient for Adequate Renal Perfusion**

Poor cardiac output
 Low systemic vascular resistance (e.g., as in sepsis)
 Volume depletion (gastrointestinal loss, poor oral intake, excess diuretic use)

Presence of Renal Vascular Disease^a

Bilateral renal artery stenosis
 Stenosis of dominant or single kidney
 Afferent arteriolar narrowing (caused by hypertension and cyclosporine)
 Diffuse atherosclerosis in smaller renal vessels

Vasoconstrictor Agents (Nonsteroidal Antiinflammatory Drugs, Cyclosporine)

^aClinical features of renal vascular disease include vascular bruits (areas of the epigastric, femoral, and carotid arteries), prior rise in serum creatinine level of more than 30%, fall in estimated glomerular filtration rate of more than 20% after beginning of treatment with an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, and a history of flash pulmonary edema.

of higher rates of hyperkalemia after publication of results of the Randomized Aldactone Evaluation Study (RALES), in which spironolactone was used in addition to ACE inhibitors for heart failure.^{224,225} The risks associated with hyperkalemia and hypokalemia have been investigated in a large meta-analysis that combined data from 27 general population and CKD cohort studies to include 1,217,986 participants. The lowest risk of adverse outcomes was observed when serum potassium was in the range 4.0 to 4.5 mmol/L. Compared with a reference value of 4.2 mmol/L, serum potassium concentration greater than 5.5 mmol/L or less than 3.0 mmol/L was associated with significantly increased risk of all-cause mortality (HR, 1.22; 95% CI, 1.15–1.29 and HR, 1.49; 95% CI, 1.26–1.76, respectively). A similar pattern was observed for the outcomes of cardiovascular mortality and ESKD.²²⁶ Randomized trials are required to establish whether correction of abnormal serum potassium is associated with improved outcomes. The discontinuation of potassium supplements, avoidance of potassium-sparing diuretics, and dietary advice to avoid high-potassium foods may all help reduce the incidence of hyperkalemia. Trials are underway to evaluate the use of novel oral potassium binding drugs to prevent hyperkalemia and facilitate RAAS inhibitor treatment for persons with CKD.¹⁸⁰

STRATEGY FOR MAXIMAL RENOPROTECTION: AIMING FOR REMISSION OF CHRONIC KIDNEY DISEASE

The earlier that interventions to slow CKD progression start, the more kidney function there is to protect and preserve. The best chance of achieving maximal renoprotection is therefore

when therapy is established as early as possible, preferably in stage 1 or 2 disease. The concepts of “remission” and “regression” have been applied to renoprotection. “Remission” indicates that therapy has been optimized to the point that there is no evidence of active disease and GFR declines by no more than expected with aging. “Regression” implies that there is recovery of renal function with improving GFR.

The fact that remission of kidney disease can be achieved was demonstrated in one of the first follow-up studies from the Captopril Collaborative Study Group in 1994.²²⁷ Of the 409 persons recruited into that study, 108 had nephrotic-range proteinuria (urine protein levels >3.5 g/day) at study entry. Remission of nephrotic-range proteinuria occurred in seven of 42 persons randomly assigned to receive captopril (16.7%) and in one of 66 persons randomly assigned to receive placebo (1.5%). Of importance is that over the follow-up period, those achieving remission had the largest initial fall in mean urine protein levels (from 5.0 g/day to 0.9 g/day, in contrast to 6.2 g/day to 5.1 g/day in participants who did not reach remission), lower SBP (135–119 mm Hg in the persons in remission; 145–143 mm Hg in those not in remission), and stable serum creatinine levels (baseline vs. final serum creatinine measurement: 1.5–1.6 mg/dL for persons in remission; 1.5–3.2 mg/dL for those not in remission). Similarly, Hovind and colleagues²²⁸ reported remission in 31% and regression (in this case, defined as GFR decline similar to that in normal aging) in 22% of 301 consecutive persons with type 1 diabetes and nephropathy who were monitored with annual measurements of isotopic GFR for 7 years. Hovind and colleagues also reported increasing prevalence of remission and regression with lower achieved blood pressure (Fig. 59.12).

Nevertheless, it is clear that even in the setting of a randomized trial, treatment with the most active agents (RAAS inhibitor therapy) is no guarantee that remission will occur; each renoprotective intervention discussed slows the rate of CKD progression by 50% at best. To achieve maximal long-term renoprotection, it is therefore necessary to use a comprehensive strategy with multiple interventions directed at different aspects of the pathogenesis of progressive renal injury (Fig. 59.2).^{229–231} Moreover, once treatments have been introduced, frequent monitoring of blood pressure, proteinuria, and GFR is essential so that therapy can be escalated until therapeutic goals have been achieved (see Tables 59.2 and 59.4). In this regard, our approach is analogous to that applied in modern oncology chemotherapeutic strategies, in which multiple agents are used and treatment is directed toward correcting all signs of disease activity until the patient is said to be in “remission.”

Data from a small number of persons suggest that if remission can be maintained over the long term, some recovery of kidney function or regression of kidney disease may be achieved.¹⁴⁰ Limited data indicate that significant improvements in renoprotection can be achieved with this strategy. Among 160 persons with type 2 diabetes and microalbuminuria, intensive therapy resulted in a marked reduction in the risk of overt nephropathy (OR, 0.27; 95% CI, 0.1 to 0.75)²²⁹ as well substantial reductions in cardiovascular events and survival as well as progression of CKD during long term follow-up for up to 21 years.^{230–233} Similarly, 26 of 56 persons with resistant nephrotic-range proteinuria and CKD who were referred to a “remission clinic” achieved reduction of urine

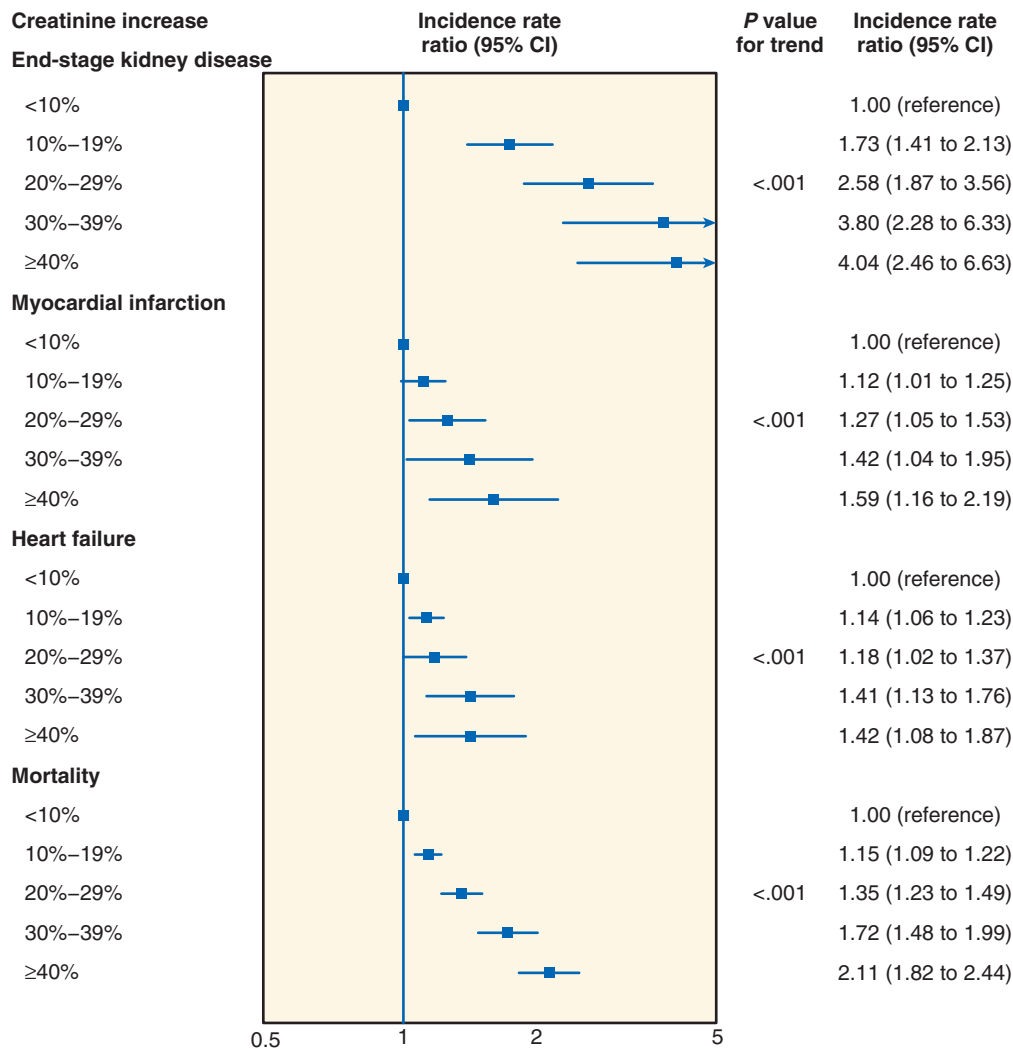


Fig. 59.11 Cardiorenal risks associated with levels of serum creatinine increase within 2 months after initiating treatment with renin-angiotensin-aldosterone system inhibitors. *CI*, Confidence interval. (From Schmidt M, Mansfield KE, Bhaskaran K, et al. Serum creatinine elevation after renin-angiotensin system blockade and long term cardiorenal risks: cohort study. *BMJ*. 2017;356:j791.)

protein levels to less than 1 g/day and stabilization of renal function after application of a similar intensive therapy protocol. Furthermore, the rate of GFR decline was significantly slower than that observed in 56 matched historical controls, and only 3.6% reached ESKD, in contrast to 30.4% of the controls.²³⁴ This strategy is based on currently available interventions, and the measurements required for monitoring are already widely used. Thus a comprehensive approach to renoprotection is an achievable goal for all persons with CKD. Although it has been argued that there is a need for new renoprotective agents, it is also true that these available therapies are not yet applied to all persons with CKD.^{235,236} If widely implemented, a comprehensive renoprotective strategy may not only delay the need for dialysis in many persons but may also substantially reduce the number of persons progressing to ESKD.

The different incidence rates of ESKD across the similarly developed nations, Norway and the United States,²³⁷ despite similar CKD prevalence rates, suggest that lower rates of progression to ESKD is a realistic goal of CKD treatment. The risk for progression in general was higher for US white

persons than for Norwegians. Of a population of 100,000 with stage 3 or 4 CKD, 610 progressed to ESKD in the United States and 240 in Norway, a 2.5-fold excess that, after adjustment, was twofold higher among nondiabetic US persons and 2.8-fold higher among diabetic US persons with stages 3 and 4 CKD. In addition, white US persons were referred later to a nephrologist. These results also underscore the need for public health systems to target or screen populations at high risk. In the United Kingdom, a pay-for-performance system for primary-care family physicians has had dramatic impacts on the screening and management of diabetes; both levels of care for diabetes and testing rates for nephropathy have increased.^{238,239} In addition, pay for performance was associated with improved control of blood pressure in persons with CKD.²⁴⁰ Regrettably this system has subsequently been discontinued. Patient self-management is a further strategy that should be pursued, as highlighted by an RCT showing improved blood pressure control in participants with hypertension and cardiovascular disease, diabetes, or CKD randomized to self-management versus usual care.²⁴¹ Reports of a decline in the incidence of new persons starting

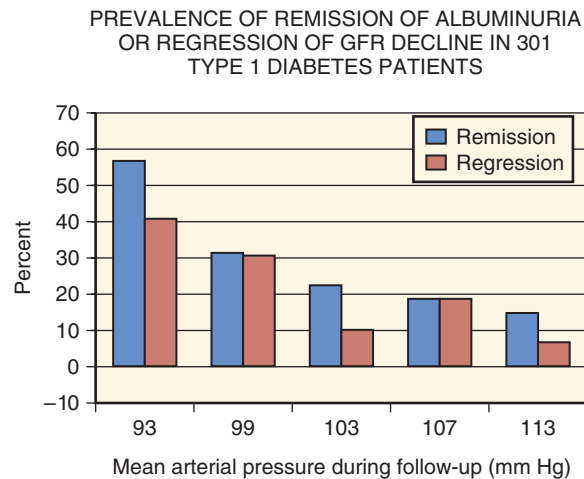


Fig. 59.12 Results from a prospective observational cohort study in which 301 consecutive persons with type 1 diabetes and diabetic nephropathy were monitored with isotopic glomerular filtration rate (GFR) measurements annually and who underwent aggressive lowering of blood pressure. Thirty persons (10%) remained normotensive (blood pressure <140/90) during this period and did not receive prolonged antihypertensive agents. 17% of the persons received monotherapy, 47% received two agents, 30% three agents, and 6% four or more agents. “Remission” was defined in this study as urine albumin levels lower than 200 µg/min sustained for at least 1 year and a decrease of at least 30% from levels before remission (surrogate endpoint). “Regression” was defined as a rate of decline in GFR equal to that in the natural aging process: less than 1 mL/min/year during the entire observation period (principal endpoint). Remission and regression may have occurred in the same persons. (Modified from Hovind P, Rossing P, Tarnow L, et al. Remission and regression in the nephropathy of type 1 diabetes when blood pressure is controlled aggressively. *Kidney Int.* 2001;60:277–283.)

dialysis therapy in the United States indicate that strategies to improve renoprotection may be starting to have an effect (Table 59.4).^{242,243}

INTERVENTIONS TO REDUCE CARDIOVASCULAR RISK ASSOCIATED WITH CHRONIC KIDNEY DISEASE

Following a seminal report in the early 1970s describing accelerated atherosclerosis in persons receiving maintenance hemodialysis,²⁴⁴ a large body of research has focused on the relationship between CKD and cardiovascular disease. It is now clear that increased cardiovascular mortality is expressed early in the course of CKD, for example, in the setting of microalbuminuria in persons with diabetes²⁴⁵ and when the GFR starts to decline²⁴⁶ in stage 3 CKD. Furthermore, the rate of CVEs associated with CKD is substantial, so that reduced eGFR is now regarded as an effective marker (or “risk equivalent”) of premature cardiovascular disease, similar to diabetes. For this reason, any strategy for the management of CKD should also include interventions to reduce the associated risk of cardiovascular disease.

The pathogenesis and management of cardiovascular disease associated with CKD are discussed in detail in Chapter 54 and are therefore reviewed only briefly here. As discussed

Table 59.4 Comprehensive Strategy and Therapeutic Goals for Achieving Maximal Renoprotection in Persons With Chronic Kidney Disease

Intervention	Goals
ACE inhibitor or ARB treatment	Urine protein level <0.5 g/day
Additional antihypertensive therapy	GFR decline <1 mL/min/year BP <130/80 mm Hg if urine albumin excretion > 30 mg/day BP <140/90 mm Hg if urine albumin excretion < 30 mg/day
Weight loss if patient is obese	Aim for 5% weight loss
Dietary salt restriction	<5 g/day (equivalent to 90 mEq sodium/day)
Dietary protein restriction	Avoid high protein intake >1.3 g/kg/day In stages 4 and 5 CKD, consider reducing intake to 0.8 g/kg/day
Tight glycemic control	HbA _{1c} <7.0% unless considered high risk for hypoglycemia
Smoking cessation	Complete cessation
Lipid-lowering therapy	Total cholesterol <200 mg/dL (5.2 mmol/L) LDL cholesterol <100 mg/dL (2.6 mmol/L)

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; CKD, chronic kidney disease; GFR, glomerular filtration rate; HbA_{1c}, hemoglobin A_{1c}; LDL, low-density lipoprotein.

throughout the text earlier, the interventions recommended for optimal renoprotection also contribute to reducing the risk of adverse CVEs. Thus lifestyle measures including smoking cessation, weight loss, and dietary sodium restriction afford both renal and cardiovascular protection. In addition, the treatment of hypertension with RAAS inhibitors and a lower blood pressure target of 130/80 mm Hg, central to achieving renoprotection, also reduces cardiovascular risk. One exception to the principle that most interventions afford renal and cardiovascular protection is treatment with statins. The largest RCT to date found a 17% RR reduction in atherosclerotic CVEs in participants randomly assigned to statin (with or without ezetimibe) therapy²⁴⁷ but no benefit with respect to CKD progression.²⁴⁸ The 2013 KDIGO guideline for lipid management in CKD recommends treating all persons aged 50 years and older with CKD not treated with dialysis or renal transplant with a statin or statin and ezetimibe, regardless of low-density lipoprotein cholesterol levels. For persons with CKD younger than 50 years, the guideline recommends similar treatment for persons with a 10-year risk of CVE greater than 10%.²⁴⁹

Clinical Relevance

All persons with chronic kidney disease should be regarded as having a high risk of cardiovascular disease and management offered to reduce this risk, including treatment with a statin.

INTERVENTIONS TO MANAGE COMPLICATIONS OF CHRONIC KIDNEY DISEASE

ANEMIA

The anemia of CKD results from a combination of reduced renal erythropoietin production (a presumed reflection of the reduction in functioning renal mass), shortened red blood cell survival, and functional iron deficiency. Anemia—defined as a hemoglobin count lower than 13 g/dL in men and lower than 12 g/dL in women⁵—can develop well before the onset of uremic symptoms. Among over 15,000 participants in the Third National Health and Nutrition Examination Survey (NHANES III), the prevalence of anemia increased from 1% at an eGFR of 60 mL/min/1.73 m² to 9% at an eGFR of 30 mL/min/1.73 m² and from 33% to 67% at an eGFR of 15 mL/min/1.73 m².²⁵⁰

If left untreated, the anemia of CKD is associated with several adverse effects. These include deterioration in cardiac function, decreased cognition and mental acuity, and fatigue. In cross-sectional studies, associations between anemia and an increased risk of morbidity and mortality, caused principally by cardiac disease and stroke, have been described in persons on dialysis.²⁵¹ In addition, anemia may influence the progression of CKD. The effects of anemia on glomerular hemodynamics have been studied in various rat models of CKD.²⁵² In each of these models, anemia was associated with a reduction in hydraulic pressure in the glomerular capillaries (P_{GC}) and amelioration of glomerulosclerosis. By contrast, prevention of anemia in the remnant kidney model by administration of erythropoietin resulted in increased systemic and glomerular blood pressures and markedly increased glomerulosclerosis (see Chapter 51 for more detailed discussion).

Despite the apparently favorable hemodynamic effects of anemia in experimental models of CKD, some human studies suggest that anemia may in fact accelerate CKD progression. In persons with inherited hemoglobinopathies, chronic anemia is associated with glomerular hyperfiltration that eventuates in proteinuria, hypertension, and ESKD.²⁵³ In several longitudinal studies, lower hemoglobin value was identified as a risk factor for CKD progression^{254,255} and ESKD.²⁵⁶ Whether this reflects more severe kidney disease in persons with lower hemoglobin values or more rapid progression secondary to low hemoglobin value and oxygen carriage per se is unclear.

Two small randomized studies revealed renoprotective benefit when anemia was corrected with erythropoietin.^{257,258} By contrast, enthusiasm for the normalization of hemoglobin has been tempered by the results of several large RCTs that revealed no benefit or adverse effects. In two studies in which effect on left ventricular mass was the primary endpoint, as well as the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT),²⁵⁹ there was no effect of higher versus lower hemoglobin target on rate of decline in GFR.^{260,261} Moreover, in the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, achievement of a higher hemoglobin target (13–15 mg/dL) was associated with a shorter time to initiation of dialysis than was achievement of the lower target (10.5–11.5 mg/

dL).²⁶² Further concern was provoked by serious adverse effects associated with higher hemoglobin targets, including increased rate of mortality²⁶³ and increased risk of stroke.²⁵⁹ Current KDIGO recommendations are therefore to treat symptomatic anemia in CKD with erythropoietin or iron supplementation, or both, to partially correct the hemoglobin and achieve a range of 10 to 11.5 mg/dL.²⁶⁴ Hemoglobin value should not exceed 13 g/dL.²⁶⁴ The pathogenesis and treatment of anemia in CKD are discussed further in Chapter 55.

MINERAL AND BONE DISORDER

Numerous cohort studies have shown strong associations between disorders of mineral metabolism and fractures, cardiovascular disease, and mortality (reviewed by the KDIGO CKD-MBD work group²⁶⁵ (see also Chapter 53). Some of caveats are relevant here. Most of the studies reporting observational data and relationships between individual parameters and clinical outcomes have been performed in dialysis populations. The limited data regarding the prevalence of biochemical and hormonal abnormalities in stages 3 to 5 CKD often do not include analyses by primary disease, which can influence the natural history of mineral and bone disorder.

Changes in mineral metabolism and bone structure are detectable much earlier in CKD than had previously been considered. There is a slow decline in levels of 1,25-dihydroxyvitamin D and 25-hydroxyvitamin D, starting once the GFR is in the range of 60 to 70 mL/min. 1,25-Dihydroxyvitamin D values were correlated positively with GFR and negatively with the log of plasma parathyroid hormone and serum phosphorus concentrations. The plasma parathyroid hormone concentration rises later in the progression of CKD but more exponentially when GFR falls below 45 mL/min/1.73 m².²⁶⁶ Calcium and phosphorus values do not generally become abnormal until GFR falls below 40 mL/min/1.73 m², and this occurs more commonly below 20 mL/min/1.73 m².²⁶⁷

In the community-based Kidney Early Evaluation Program (KEEP) and NHANES cohorts, there is evidence that parathyroid hormone levels increased early in persons with stage 3 CKD, typically while calcium and phosphorus levels remained normal.^{267,268} These findings highlight the importance of monitoring parathyroid hormone along with calcium and phosphorus in individuals with eGFR less than 60 mL/min/1.73 m². Investigators have also described an earlier biomarker of altered mineral and bone disorder in persons with CKD. Fibroblast growth factor-23 (FGF-23) regulates phosphorus metabolism and is associated with mortality in persons on dialysis.²⁶⁹ High levels of this growth factor, defined as being above 100 reference units/mL, were more common than secondary hyperparathyroidism and hyperphosphatemia at all levels of eGFR in a cohort of almost 4000 persons with stages 2 to 4 CKD.²⁷⁰ Further analysis in the same cohort identified elevated FGF-23 levels as an independent risk factor for progression to ESKD and mortality.²⁷¹ However, testing for this early marker is not routinely available, and so we recommend monitoring according to the KDIGO guidelines.²⁷²

Thus monitoring of serum levels of calcium, phosphorus, and parathyroid hormone and of alkaline phosphatase activity should begin in stage 3 CKD. Few persons with mineral and

bone disorder in CKD develop symptomatic disease until they have stage 5 CKD; therefore the main reason for monitoring is to implement early preventive treatment to suppress secondary hyperparathyroidism. In the majority of persons with CKD who do not require dialysis, treatment of mineral and bone disorder is based initially on dietary phosphate restriction, followed by oral phosphate binders in order to maintain phosphorus in the normal range, although there is no evidence from RCTs that this affects clinical outcomes.

In persons with stages 3 to 5 CKD who are not on dialysis, the optimal level of parathyroid hormone is unknown. However, it is suggested that persons with levels of intact parathyroid hormone above the upper normal limit of the assay be first evaluated for hyperphosphatemia, hypocalcemia, and vitamin D deficiency.⁵ If parathyroid hormone level is progressively rising and remains persistently above the upper limit of normal for the assay despite correction of modifiable factors, treatment with calcitriol or vitamin D analogs (once phosphate is under control) may be considered. Treatment with the vitamin D analog paricalcitol has been reported to reduce albuminuria in persons with diabetic nephropathy already receiving treatment with an ACE inhibitor or an ARB,²⁷³ but further studies are needed to investigate the renoprotective potential of vitamin D therapies. Detailed discussion of the pathogenesis and treatment of mineral and bone disorder in CKD are beyond the scope of this chapter and are presented in Chapter 53.

A STEPPED CARE APPROACH TO CHRONIC KIDNEY DISEASE

In addition to being useful for risk stratification, the KDIGO classification of CKD serves as a convenient framework to identify differing priorities for management at different stages. This “stepped care” approach is discussed in the following sections and summarized in [Table 59.2](#).

STAGES 1 AND 2

At these stages of CKD, the diagnosis is based on the presence of albuminuria, hematuria, or structural kidney disease, and an eGFR above 60 mL/min/1.73 m². Stage 1 disease is defined by a GFR greater than 90 mL/min/1.73 m² (5.7% of the total US population) and stage 2 by a GFR between 60 and 89 mL/min/1.73 m² (5.4% of the US population).²⁷⁴ These statistics indicate that CKD of all stages affected an estimated 16.8% of adults aged 20 years during the period 1999 to 2004, an increase from the recalculated NHANES III (1988–1994) estimate of 14.5%.²⁷⁵

Persons with stages 1 and 2 disease do not have specific symptoms or complications of renal failure such as anemia or bone and mineral disorder. Those with symptoms may have a multisystem disease with secondary glomerular or interstitial disease (see Chapters 32 and 35). The majority of persons with stages 1 and 2 CKD are detected by routine or health care insurance-mandated screening and are visiting primary care or other physicians; in many countries only a small proportion are ever evaluated by a nephrologist. With increased access to radiologic imaging, more persons are now identified with structural abnormalities such as polycystic kidney disease or single kidney.

The emphasis at these early stages should be on identification of specific renal diseases when present, appropriate referral to a nephrologist, and reduction of cardiovascular risk. A detailed family history is important because persons with a positive family history need more detailed investigation to allow early detection of inherited renal disease and, in particular, adult polycystic kidney disease. The following initial investigations are appropriate for assisting with risk assessment and for informing decisions about referral to a nephrologist or urologist (see also Chapter 23):

1. Estimation of urinary albumin or protein excretion by measurement of ACR or PCR on a random urine sample. Persons with a urine protein measurement equivalent to 0.5 g/day (UACR, 300 mg/g or 30 mg/mmol) or greater should be referred for investigation by a nephrologist.⁵
2. Further urinalysis is needed to detect hematuria. For painless but visible hematuria, serious urologic causes—such as bladder, renal cell, and, less often, prostatic cancers—must be confirmed or ruled out, particularly in persons older than age 50 years, smokers, and those with a family history of renal tract malignancy. They should generally be assessed by a urologist or nephrologist who has experience in screening for these conditions. Painless microscopic hematuria (nonvisible hematuria) is much more likely to be caused by glomerular disease, but referral to a urologist may be necessary to confirm or rule out renal tract malignancy in persons at increased risk.²⁷⁶ There are several national guidelines for the investigation of hematuria.^{276,277} Initial blood tests include measurements of creatinine, eGFR, urea, electrolytes (Na⁺, K⁺, HCO₃⁻, Cl⁻), bone and liver profiles, blood glucose, glycosylated hemoglobin, blood cell count, erythrocyte sedimentation rate, and (in men) prostate-specific antigen. Serologic screening for underlying myeloma, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, antglomerular basement membrane disease, and systemic lupus erythematosus may be indicated, depending on symptoms and the results of other investigations.
3. Abdominal ultrasonography to exclude structural abnormalities and determine the bipolar diameter of the kidneys is indicated if urinalysis results are abnormal, if there is a strong family history of CKD, or if there is significant hypertension. Asymmetry with regard to renal size may be suggestive of atherosclerotic renovascular disease, and angiography (computed tomography or magnetic resonance angiography) may therefore be helpful if this is suspected.

In general, persons with stages 1 and 2 CKD, who do not have a specific renal disease or significant proteinuria, require only annual monitoring of blood pressure, eGFR, and proteinuria. Those who develop an abrupt or sustained decline in GFR or worsening proteinuria should be referred to a nephrologist for further investigation and optimization of therapy.⁵

STAGE 3

In stage 3 CKD, GFR is between 30 and 59 mL/min/1.73 m². This is a significant stage because it represents the majority of persons in whom CKD is identified (stages 1 and 2 often

Table 59.5 Recommended Frequency of Monitoring by Stage of Chronic Kidney Disease

Variable	Stage 1 and 2	Stage 3	Stage 4	Stage 5
GFR and electrolytes	Every 12 months	Every 3–12 months	Every 3–6 months	Every 1–3 months
Proteinuria with ACR or PCR testing	Every 12 months	Every 3–12 months	Every 3–6 months	Every 3–6 months
Blood pressure	Each visit	Each visit	Each visit	Each visit
Calcium and phosphate levels	Every 12 months	Every 12 months	Every 3–6 months	Every 3 months
Parathyroid hormone level	—	Every 12 months	Every 3–6 months	Every 3–6 months ^a
Hemoglobin	Every 12 months	Every 12 months	Every 3–6 months	Every 1–3 months ^a

Monitoring should be individualized according to previous rate of GFR decline, risk assessment of future GFR decline (particularly high if heavy proteinuria, >1 g/day or equivalent), and current drug therapy.

^aMonitoring of parathyroid hormone and anemia should depend on the previous results and specific treatment, if any, for these conditions. Stable values with no specific treatment require less monitoring as indicated.

ACR, Albumin-to-creatinine ratio; GFR, glomerular filtration rate; PCR, protein-to-creatinine ratio.

remain undetected unless urinalysis is performed) and because many of the complications start to manifest once the GFR drops below 45 mL/min/1.73 m². In addition, the rate of cardiovascular mortality increases substantially among persons with a GFR lower than 45 mL/min/1.73 m².^{246,278} Some national guidelines and the KDIGO guidelines have therefore split this stage into two: 3A, defined by reduced GFR of 45 to 59 mL/min/1.73 m², and 3B, defined by reduced GFR of 30 to 44 mL/min/1.73 m².^{5,279,280} Depending on the underlying etiology, Stage 3 CKD may be appropriately managed by a collaboration of primary care with a nephrology service. One study that employed mathematical modeling techniques reported that multidisciplinary care for CKD is a cost-effective model that could help to reduce the need for dialysis and prolong life expectancy.²⁸¹ The aims of management at this stage are to identify specific renal disease, correct reversible causes of renal dysfunction, prevent or slow the progression of CKD, reduce cardiovascular risk, and treat the complications of CKD (usually in stage 3B CKD). Referral criteria are the same as for earlier stages of CKD, with the addition of anemia and mineral and bone disorder, which may necessitate specialist treatment. Monitoring of blood pressure, eGFR, and serum biochemistry profile, as well as complete blood cell count and evaluation for proteinuria, should be performed every 3 to 12 months, depending on risk profile and clinical circumstances (Table 59.5).

STAGE 4

Persons with stage 4 CKD have a high cumulative risk of cardiovascular death and progression to ESKD. Almost 66% of such persons experience either a renal event or a CVE over 5 years after diagnosis. In a population-based study,²⁸² the proportions of persons who needed RRT during a 5-year observation period were 1.1%, 1.3%, and 19.9%, respectively, for stages 2, 3, and 4, and the respective mortality rates were 19.5%, 24.3%, and 45.7%. Not surprisingly, persons with stage 4 disease often make up a large proportion of those attending outpatient nephrology clinics. Achieving renoprotection remains an important goal to delay the onset of RRT for as long as possible, as does minimizing cardiovascular risk. Blood pressure, eGFR, and serum biochemistry profile, including level of parathyroid hormone as well as complete blood cell count, should be monitored every 3 to 6 months.

As the GFR declines to below 20 mL/min/1.73 m², the focus should change to treating the complications of CKD and planning for RRT.²⁸³ Effective preparation for RRT requires input from multiple staff disciplines (medical, nursing, pharmacy, dietetics, psychology, and social work) and is best delivered in a multidisciplinary clinic. There is emerging evidence that persons prefer this approach to preparation and that such clinics are associated with better outcomes, at least in observational studies.^{284,285}

It is clear that late referral (less than 3 months before initiation of dialysis) for dialysis preparation is associated with significantly higher rates of mortality²⁸⁶ and lower quality of life.²⁸⁷ Results of a Canadian study also indicated that even when referral was appropriately timed, there was a 53% higher rate of reaching the composite endpoint of death, need for transfusion, or subsequent hospitalization in those without an optimal start to RRT.²⁸⁸

The need for timely preparation for dialysis is clear and is emphasized in the majority of national guidelines. However, not all persons with stage 4 CKD progress to stage 5, and unnecessary preparation may do harm. Persons with stage 4 CKD should therefore undergo a formal assessment of their risk of progression to ESKD (see Chapter 20). This is particularly relevant in older persons, in whom the competing risk of death is an important consideration. A comprehensive approach to assessing the risk of ESKD versus death to assist decision-making about RRT has been proposed by the European Renal Best Practice (ERBP) group.²⁸⁹ Preparation for initiation of dialysis requires multiple interventions to deal with both medical and psychosocial aspects. Persons require adequate counseling to assist them in the choice of dialysis modality and in coping with the psychosocial effects of starting dialysis. Elderly persons are often more accepting of dialysis than are younger persons, who may still be working or have family commitments. One large US study confirmed that social support is important for persons on hemodialysis and peritoneal dialysis in terms of greater satisfaction, higher quality of life, and fewer hospitalizations.²⁹⁰ Timely formation of vascular access, ideally a forearm arteriovenous fistula, is important to allow adequate maturation, modification if necessary, and repeat surgery in case of primary site failure. Peritoneal catheter insertion requires less maturation time but should be performed early enough to allow time for adequate training for peritoneal dialysis. See Chapters 63,

64, and 68 for further discussion of preparation for dialysis and vascular access.

HEPATITIS B VACCINATION

Persons on hemodialysis have a small but significantly increased risk of exposure to hepatitis B and other blood-borne viruses. Severe outbreaks of hepatitis B in hemodialysis units have resulted in considerable morbidity and even mortality among susceptible persons and staff. Therefore persons with CKD in whom dialysis is anticipated should be screened for hepatitis B and C, as well as human immunodeficiency virus infection. Persons who are seronegative for hepatitis B surface antigen and hepatitis B surface antibody should be immunized and their antibody levels measured after vaccination. Because seroconversion rates decrease with GFR,²⁹¹ immunization should ideally occur in stage 3 in persons with a high risk of progression; however, in view of the large number of persons and lack of precision in predicting outcomes, it is usually delayed until stage 4. Seroconversion rates are low once dialysis has commenced, particularly in older persons.²⁹² Results of a meta-analysis of 12 studies indicated that increased seroconversion rates can be achieved by administering multiple doses of vaccine and preferably by the intradermal route.²⁹³ The mechanism is unclear, but this finding is consistent with the observed increased immunogenicity of other vaccinations after intradermal administration in persons with CKD and the rare cases of pure red blood cell aplasia after subcutaneous administration of recombinant erythropoietin.^{294,295}

PREEMPTIVE RENAL TRANSPLANTATION

Assessment and preparation for possible kidney transplantation should be undertaken before initiation of dialysis. The increase in death rates among waitlisted persons in comparison with transplant recipients is consistent although still debated in view of methodologic issues, such as lead-time bias and unmeasured differences confounding these analyses.²⁹⁶ The clear benefits to recipients and the potential reduced demand for dialysis make a strong case for strategies to increase preemptive transplantation. Many countries and centers now permit kidney transplantation when the recipient's GFR is less than 15 to 20 mL/min/1.73 m² and if renal function has decreased progressively over the previous 6 to 12 months.²⁹⁷ The optimal timing of preemptive kidney transplantation remains unclear. Neither higher nor lower pretransplantation GFR appears to be associated with superior allograft survival.²⁹⁸

STAGE 5

Once GFR declines to below 15 mL/min/1.73 m², priorities include maintaining optimal health and function as well as achieving a planned and uncomplicated initiation of RRT. If persons have been referred in a timely manner, preparation for RRT should already be complete, but ongoing psychosocial support is often necessary as persons come to terms with the imminent need to start therapy. The optimal time to initiate RRT remains a topic of debate.²⁹⁹ A retrospective analysis of data from 896,546 persons commencing dialysis revealed a higher mortality rate among those commencing dialysis “early” (GFR >15 mL/min/1.73 m²) and a lower mortality rate among those commencing “late” (GFR

<5 mL/min/1.73 m²), although the authors conceded that the results may have been affected by unaccounted confounding and selection, as well as lead-time bias.³⁰⁰ In a landmark RCT, persons with stage 5 CKD were randomly assigned to “early” (GFR = 10–14 mL/min/1.73 m²) or “late” (GFR = 5–7 mL/min/1.73 m²) initiation of dialysis. After a median of 3.59 years of follow-up, there was no difference in rate of survival or adverse events (CVEs, infections, or complications of dialysis) between the groups.³⁰¹

We therefore recommend that the initiation of RRT should be individualized but in general should occur when the GFR falls below 10 mL/min/1.73 m² but before significant uremic symptoms or malnutrition occurs. In order to facilitate this timing, the frequency of monitoring of GFR, serum biochemistry, and hemoglobin, together with clinical assessment, should increase to every 1 to 3 months. Persons who decline RRT should continue to be treated for complications of CKD to optimize their quality of life and, if necessary, be referred to a palliative care service to allow adequate planning of their care once they develop symptomatic uremia.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Which of the following statements regarding the treatment of hypertension in chronic kidney disease is true?
- Randomized controlled trials have provided clear evidence to guide recommendations for blood pressure targets.
 - The treatment of hypertension is effective in slowing chronic kidney disease (CKD) progression.
 - Treatment with a direct renin inhibitor is more renoprotective than treatment with an angiotensin receptor blocker (ARB).
 - Combination treatment with an angiotensin-converting enzyme inhibitor and ARB is recommended.
 - Current guidelines all recommend the same blood pressure targets.

Answer: b

Rationale: Though several trials and a large meta-analysis confirmed that the treatment of hypertension is effective in slowing CKD progression, trials that have investigated specific lower blood pressure targets have generally produced equivocal results. As a result, current guidelines recommend slightly different blood pressure targets, although recently there is more consensus recommending a target of less than 130/80 mm Hg for all persons with CKD (see the “[Antihypertensive Therapy](#)” section).

2. Treatment with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) has not been shown to provide renoprotective benefit in which of the following categories of people with chronic kidney disease (CKD)?
- Type 1 diabetes with albuminuria
 - Type 2 diabetes with albuminuria
 - Nondiabetic CKD with albuminuria
 - Diabetic and nondiabetic CKD without albuminuria
 - Nondiabetic CKD with proteinuria >3 g/day

Answer: d

Rationale: Evidence of renoprotective benefit from treatment with ACE inhibitors or ARBs has been clearly established by randomized controlled trials in all forms of CKD with albuminuria, but not in the absence of albuminuria (see the “[Pharmacologic Inhibition of the Renin–Angiotensin–Aldosterone System](#)” section).

3. Using the 2012 Kidney Disease Improving Global Outcomes (KDIGO) chronic kidney disease (CKD) classification system, identify the correct category for a patient with an estimated glomerular filtration rate of 42 mL/min/1.73 m² and urine albumin-to-creatinine ratio of 400 mg/g.
- G3a A3
 - G3a A2
 - G3b A1
 - G3b A2
 - G3b A3

Answer: e

Rationale: See the “[Classification of CKD](#)” section.

4. Which of the following treatments is not recommended in the 2012 Kidney Disease Improving Global Outcomes (KDIGO) guidelines for the management of chronic kidney disease (CKD)?
- Angiotensin-converting enzyme (ACE) inhibitor for patients with albuminuria
 - Aspirin for all people with CKD
 - Sodium bicarbonate for patients with serum bicarbonate <22 mEq/L
 - Combination ACE inhibitor and angiotensin receptor blocker (ARB) therapy
 - b and d

Answer: c

Rationale: There is uncertainty regarding the risks versus benefits of aspirin in persons with CKD and this is therefore not recommended in the guidelines. Combination of ACE inhibitor and ARB therapy is not recommended due to data from randomized controlled trials showing no additional benefit versus monotherapy and an increase in risk of acute kidney injury as well as hyperkalemia (see the “[Pharmacologic Inhibition of the Renin–Angiotensin–Aldosterone System](#)” section).

5. Which of the following statements regarding lipid-lowering therapy in chronic kidney disease (CKD) is true?
- Statin treatment is contraindicated in people on dialysis.
 - Statin treatment is renoprotective.
 - Lipid-lowering therapy reduced atherosclerotic cardiovascular events in a large randomized controlled trial (RCT) of patients with CKD.
 - Statins are contraindicated in people with CKD stage 5.
 - Statins do not lower cholesterol in people with CKD.

Answer: c

Rationale: A large randomized controlled trial (SHARP) reported a 17% relative risk reduction in atherosclerotic cardiovascular events in persons with CKD randomized to treatment with simvastatin with or without ezetimibe but no benefit with respect to renal outcomes. The analysis indicated no difference in effects between participants receiving dialysis versus not on dialysis, though other trials have reported a lack of benefit with statin treatment in persons receiving dialysis (see the “[Interventions to Reduce Cardiovascular Risk Associated with Chronic Kidney Disease](#)” section).